

Probing the role of synaptic adhesion molecule RTN4RL2 in setting up cochlear connectivity

Reviewed Preprint

v2 • May 23, 2025

Revised by authors

Reviewed Preprint

v1 • February 12, 2025

Nare Karagulyan, Maja Überegger, Yumeng Qi, Norbert Babai, Florian Hofer, Lejo Johnson Chacko, Fangfang Wang, Maria Luque, Rudolf Glueckert, Anneliese Schrott-Fischer, Yunfeng Hua , Tobias Moser , Christine Bandtlow 

Institute for Auditory Neuroscience, University Medical Center Göttingen, Göttingen, Germany • Auditory Neuroscience and Synaptic Nanophysiology Group, Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany • Cluster of Excellence “Multiscale Bioimaging of Excitable Cells”, Göttingen, Germany • Institute of Neurobiochemistry, Biocenter, Medical University of Innsbruck, Innsbruck, Austria • Shanghai Institute of Precision Medicine, Ninth People’s Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China • Department of Otorhinolaryngology, Medical University of Innsbruck, Innsbruck, Austria • Department for Hearing, Speech and Voice Disorders, Medical University of Innsbruck, Innsbruck, Austria

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

In this work, the authors characterize the synaptic adhesion molecule RTN4RL2, demonstrating its critical involvement in the development and function of auditory synapses between inner hair cells and spiral ganglion neurons. This study is **important** because it offers potential insights into therapeutic strategies for hearing loss associated with synaptic dysfunction. The findings are **solid**, because they are supported by the use of multiple advanced techniques, including FISH and SBEM imaging.

<https://doi.org/10.7554/eLife.103481.2.sa3>

Abstract

Sound encoding depends on the precise and reliable neurotransmission at the afferent synapses between the sensory inner hair cells (IHCs) and spiral ganglion neurons (SGNs). The molecular mechanisms contributing to the formation, as well as interplay between the pre- and postsynaptic components remain largely unclear. Here, we tested the role of the synaptic adhesion molecule and Nogo/RTN4 receptor homolog RTN4RL2 (also referred to as Ngr2) in the development and function of afferent IHC-SGN synapses. Upon deletion of RTN4RL2 in mice (RTN4RL2^{-/-}), presynaptic IHC active zones showed enlarged synaptic ribbons and a depolarized shift in the activation of Ca_v1.3 Ca²⁺ channels. The postsynaptic densities (PSDs) of SGNs were smaller and deficient of GluA2-4 AMPA receptor subunits despite maintained Gria2 mRNA expression in SGNs. Next to synaptically engaged PSDs we observed “orphan” PSDs located away from IHCs. They likely belong to a subset of SGN peripheral neurites that do not contact the IHCs in RTN4RL2^{-/-} cochleae as found by volume electron microscopy reconstruction of SGN neurites. Auditory brainstem responses of RTN4RL2^{-/-} mice showed

increased sound thresholds indicating impaired hearing. Together, these findings suggest that RTN4RL2 contributes to the proper formation and function of auditory afferent synapses and is critical for normal hearing.

Introduction

Hearing relies upon the correct formation and maturation of cochlear afferent synapses between postsynaptic type I spiral ganglion neurons (SGNs) and presynaptic inner hair cells (IHCs; reviewed in Johnson et al., 2019 [↗](#); Bulankina and Moser, 2012 [↗](#); Appler and Goodrich, 2011 [↗](#)). In mice, SGN neurites extend and reach IHCs at late embryonic stages (E16.5; Koundakjian, Appler, and Goodrich, 2007 [↗](#)), whereby early synaptic contacts are established between the two cells at E18 (Michanski et al., 2019 [↗](#)). Developmental changes and maturation of afferent synapses continue until the onset of hearing (p12), with further refinement occurring till the fourth postnatal week (Wong et al., 2014 [↗](#); Liberman and Liberman, 2016 [↗](#); Michanski et al., 2019 [↗](#)). In mature cochleae, each IHC receives a contact from 5-30 SGNs, while each SGN contacts only one active zone (AZ) of one IHC in the majority of cases (Meyer and Moser, 2010 [↗](#); Hua et al., 2021 [↗](#)).

To this date, multiple mechanisms have been implicated in SGN neurite guidance and establishment of IHC innervation by SGNs, including signaling via EphrinA5-EphA4, Neuropilin2/Semaphorin3F, Semaphorin5B/PlexinA1, Semaphorin3A (Defourny et al., 2013 [↗](#); Coate et al., 2015 [↗](#); Jung et al., 2019 [↗](#); Cantu-Guerra et al., 2023 [↗](#)). Less is known about the molecules governing the transsynaptic organization at IHC afferent synapses. Similar to the central conventional synapses, synaptic adhesion proteins such as neuroligins and neuroligins have been suggested to play a role in pre- and postsynaptic assemblies (Ramirez et al., 2022 [↗](#); Jukic et al., 2024 [↗](#)). Furthermore, the $\text{Ca}_v1.3$ extracellular auxiliary subunit $\text{Ca}_v\alpha_2\delta_2$ was indicated to be important for proper alignment of presynaptic IHC AZs and postsynaptic densities (PSDs) of SGNs (Fell et al., 2015 [↗](#)).

Whether reticulon 4 receptors (RTN4Rs) contribute to setting up afferent connectivity in the cochlea remained to be investigated. The RTN4 receptor family consists of 3 homologous proteins RTN4R, RTN4RL1 and RTN4RL2. RTN4Rs are leucine-rich repeat (LRR) and glycosylphosphatidylinositol (GPI) anchored cell surface receptors (**Figure 1A** [↗](#)). Their primary role is thought to limit synaptic plasticity and axonal outgrowth as well as to restrict axonal regeneration after injury (reviewed in Mironova and Giger, 2013 [↗](#)). While RTN4R and RTN4RL1 are involved in axonal guidance (Vaccaro et al., 2022 [↗](#)), RTN4RL2 has been proposed to be important for innervation of the epidermis by dorsal root ganglion neurons (Bäumer et al., 2014 [↗](#)). Furthermore, RTN4Rs are suggested to control the number and the development of synapses (Wills et al., 2012 [↗](#)) and to play a role in transsynaptic signaling (Wang et al., 2021 [↗](#)). Recent single-cell transcriptomic studies of the cochlea detected the expression of RTN4Rs in SGNs and IHCs (Shrestha et al., 2018 [↗](#); Jean et al., 2023 [↗](#)). Here, we investigated the role of RTN4RL2 in the cochlea using previously described RTN4RL2 constitutive knock-out mice (RTN4RL2^{-/-}; (Wörter et al., 2009 [↗](#))). We found RTN4RL2 to be expressed in SGNs and to be required for normal hearing: auditory brainstem responses were impaired upon RTN4RL2 deletion. We discovered both pre- and postsynaptic alterations of IHC-SGN synapses in RTN4RL2^{-/-} mice: presynaptic Ca^{2+} channels of IHCs required stronger depolarization to activate and PSDs seemed deficient of GluA2-4 AMPA receptor subunits. Additionally, a subset of type I SGN neurites did not contact IHCs but likely still feature “orphan” PSDs.

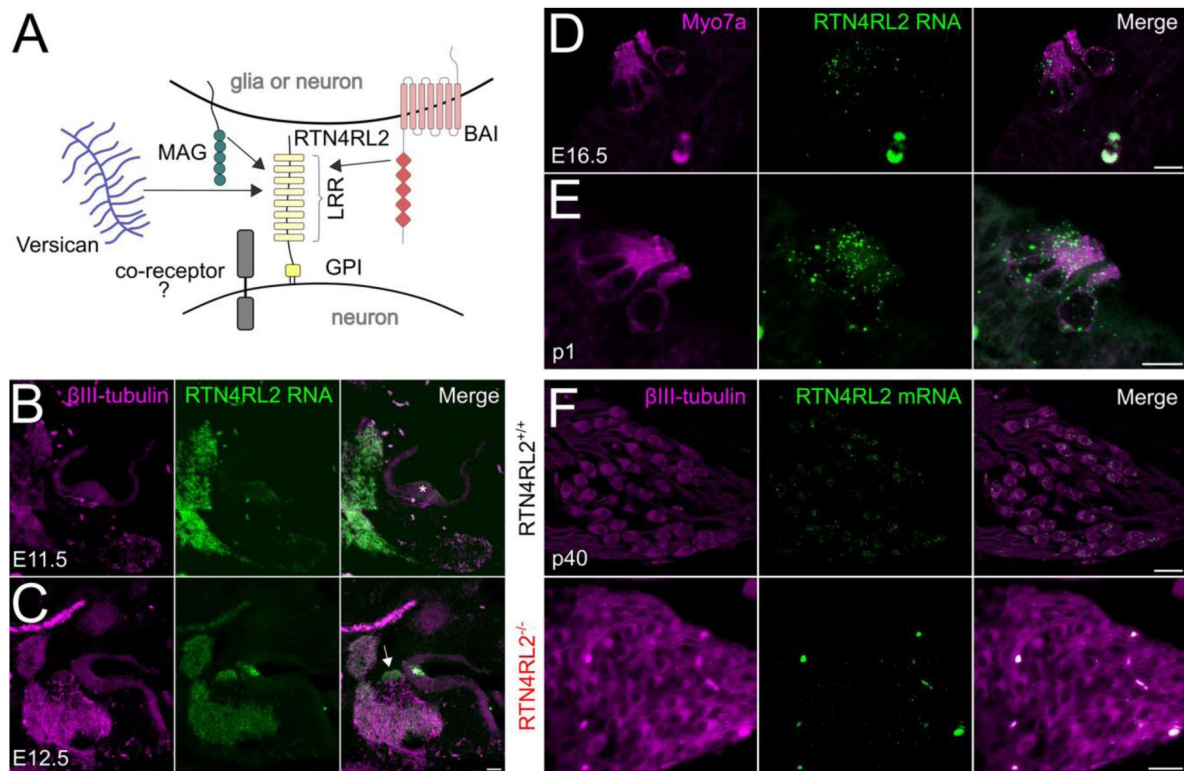


Figure 1.

RTN4RL2 mRNA expression in IHCs and SGNs of the mouse cochlea.

(A) RTN4RL2 is an LRR protein and is anchored to the cell membrane via GPI. In the nervous system RTN4RL2 has been implicated to interact with MAG, versican, BAI (Venkatesh et al., 2005; Bäumer et al., 2014; Wang et al., 2021). (B, C) Exemplary images of RNAscope ISH for RTN4RL2 mRNA (green) in the otic vesicles of E11.5 (B) and E12.5 (C) mice. Scale bar = 50 μ m. The developing organ of Corti region is marked with an asterisk and the spiral ganglion is indicated with an arrow. (D, E) Exemplary images of RTN4RL2 RNA expression in hair cells of E16.5 (D) and p1 (E) mice. Hair cells are visualized with anti-Myo7a stainings. Scale bars = 10 μ m. (F) Representative images of RNAscope ISH for RTN4RL2 mRNA (green dots) combined with immunostaining for neuron-specific marker β III-tubulin (grey) in paraffin sections of p40 RTN4RL2^{+/+} and RTN4RL2^{-/-} cochleae. Scale bars = 20 μ m.

Results

RTN4RL2 is expressed both in hair cells and spiral ganglion neurons

Recent transcriptomic data had detected RTN4RL2 expression both in hair cells and SGNs (Elkon et al., 2015 [↗](#); Liu et al., 2018 [↗](#); Shrestha et al., 2018 [↗](#); Jean et al., 2023 [↗](#)). We verified this by performing RNAscope staining in cochleae during inner ear development. We observed RTN4RL2 mRNA expression at E11.5 in the hair cell region and at E12.5 in both the spiral ganglion and hair cells (**Figure 1B** [↗](#), C). RTN4RL2 expression was further evident in hair cells at E16.5 and p1 (**Figure 1D** [↗](#), E) and was maintained at low levels in spiral ganglia of p40 mice (**Figure 1F** [↗](#)). RTN4RL2^{-/-} mice were lacking RTN4RL2-specific mRNA puncta (**Figure 1F** [↗](#)), demonstrating specific detection of RTN4RL2 expression in SGNs.

RTN4RL2 is important for the correct development of the auditory afferent synapses

To probe the role of RTN4RL2 in the cochlea, we first studied the numbers of cochlear cells in RTN4RL2^{-/-} mice. We did not observe any change in SGN density or counts of inner and outer hair cells at p15, 1-month- and 2-month-old mice (**Figure 2-figure supplement 1** [↗](#)). Given that RTN4RL2 has been implicated in synapse formation and development (Wills et al., 2012 [↗](#); Borrie et al., 2014 [↗](#); Wang et al., 2021 [↗](#)) we immunolabeled IHC afferent synapses for presynaptic RIBEYE/Ctbp2 and postsynaptic Homer1 in mice at the age of 3 weeks. While we did not detect any change in the number of synaptic ribbons in RTN4RL2^{-/-} IHCs, ribbon volumes were bigger (**Figure 2C, D** [↗](#); **Figure 2-figure supplement 2** [↗](#)). Interestingly, in addition to the Homer1 positive puncta juxtaposing presynaptic ribbons, we observed additional Homer1 patches, which appeared to be away from IHCs, potentially marking “orphan” PSDs (**Figure 2A** [↗](#), B). Moreover, the Homer1 puncta juxtaposing IHC AZs were significantly smaller in RTN4RL2^{-/-} mice compared to the control IHCs (**Figure 2E** [↗](#)). While we found the percentage of presynaptic ribbons juxtaposing Homer1 immunofluorescent puncta to be decreased by approximately 7% in RTN4RL2^{-/-} mice (**Figure 2F** [↗](#)), we cannot exclude that our immunolabeling protocol lacked the sensitivity to detect smaller synaptically engaged PSDs. Out of the four pore-forming AMPA receptor subunits (GluA1-4), mature SGNs express GluA2-4 (Niedzielski and Wenthold, 1995 [↗](#); Matsubara et al., 1996 [↗](#)). Our immunolabeling of anti-GluA2/3 revealed a severe reduction in the number of GluA2/3 positive puncta in RTN4RL2^{-/-}, indicating possible lack or decrease of GluA2/3 (**Figure 3A** [↗](#), B). Interestingly, the remaining GluA2/3 patches did not properly juxtapose presynaptic ribbons (**Figure 3C** [↗](#)). Despite this, the expression of Gria2 mRNA appeared to be maintained in SGNs, as indicated by RNAscope (**Figure 3D** [↗](#)). We then checked whether the potential lack of GluA2/3 was accompanied by an increase in GluA4 signal at the PSDs of RTN4RL2^{-/-} SGNs, as was previously shown in GluA3 KO mice (Rutherford et al., 2023 [↗](#)). Surprisingly, we did not detect a significant GluA4 signal compared to the background in RTN4RL2^{-/-} cochleae (**Figure 3-figure supplement 1** [↗](#)). To check if the “orphan” PSDs away from IHCs were possibly erroneously engaged with efferent presynaptic terminals we stained the latter for synapsin, which is lacking from IHCs, but did not observe any obvious juxtaposition between the “orphan” Homer1 and synapsin puncta (**Figure 2-figure supplement 3** [↗](#)).

Additional non-synaptic neurites in the cochlea of RTN4RL2^{-/-} mice

To further examine the afferent cochlear connectivity, we utilized serial block-face scanning electron microscopy (SBEM) to 3D reconstruct the apical cochlea segments from two RTN4RL2^{-/-} and one littermate RTN4RL2^{+/+} control mice at p36 (**Figure 4A** [↗](#)). As recently demonstrated (Hua

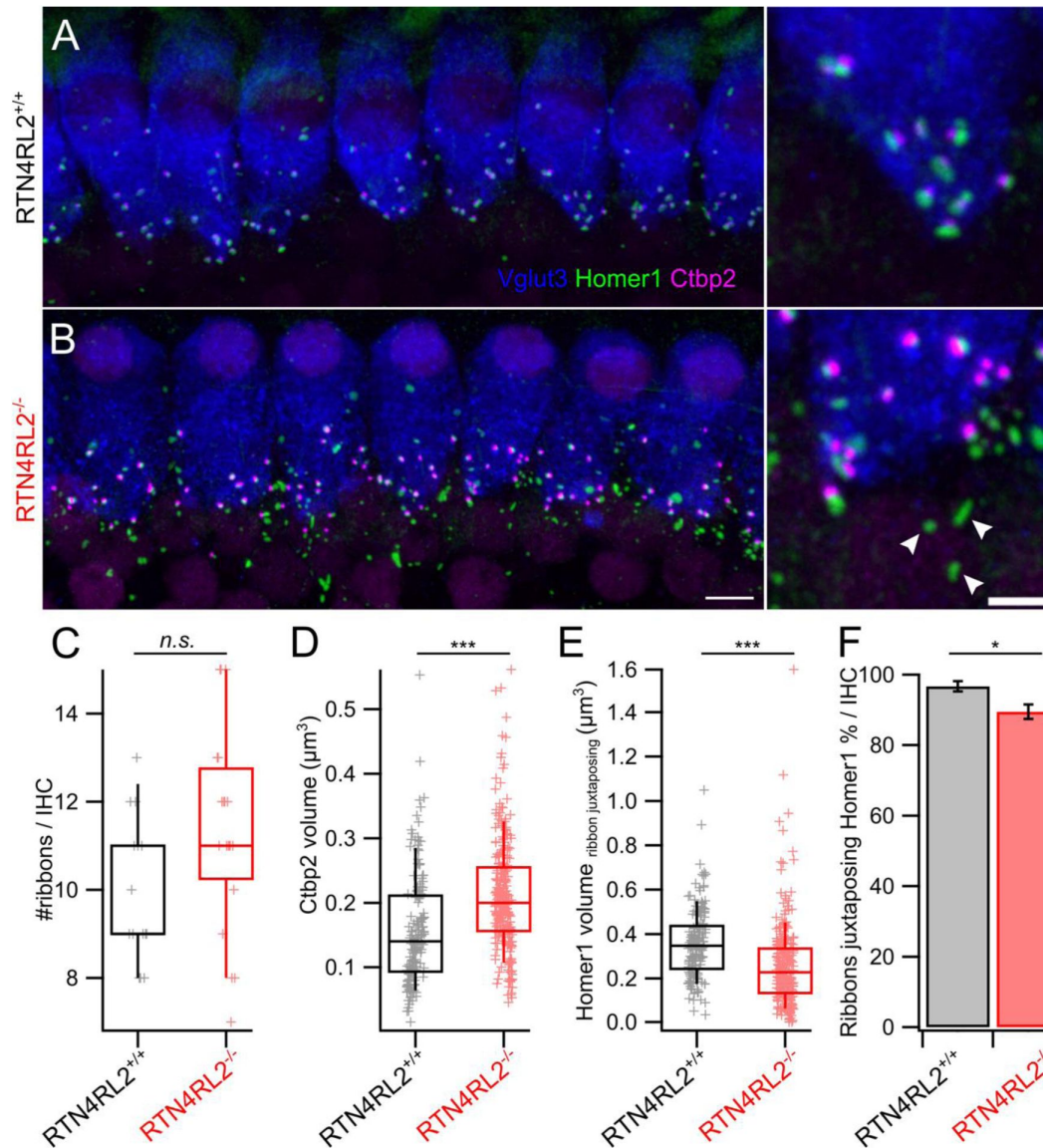


Figure 2.

Pre- and postsynaptic changes at IHCs of RTN4RL2^{-/-} mice.

(A, B) Maximum intensity projections of representative confocal stacks of IHCs from apical cochlear region of 3-week-old RTN4RL2^{+/+} (A) and RTN4RL2^{-/-} (B) mice immunolabeled against Vglut3, Homer1, and Ctbp2. Scale bar = 5 μm. Images on the right-hand side are zoomed into the synaptic regions. Scale bar = 2 μm. Some of the putative “orphan” PSDs are marked with the white arrowheads. (C) Number of Ctbp2 positive puncta is not changed in IHCs of RTN4RL2^{-/-} mice (RTN4RL2^{+/+}: 9.9 ± 0.42 , SD = 1.62, n = 15, N = 2 vs RTN4RL2^{-/-}: 11.4 ± 0.5 , SD = 2.25, n = 20, N = 3; p = 0.06, Mann-Whitney-Wilcoxon test). (D) Ribbon volumes are enlarged in RTN4RL2^{-/-} IHCs (RTN4RL2^{+/+}: $0.16 \pm 0.007 \mu\text{m}^3$, SD = $0.09 \mu\text{m}^3$, n = 165, N = 2 vs RTN4RL2^{-/-}: $0.21 \pm 0.005 \mu\text{m}^3$, SD = $0.09 \mu\text{m}^3$, n = 259, N = 3; p < 0.001, Mann-Whitney-Wilcoxon test). (E) Homer1 patches which are juxtaposing presynaptic ribbons show decreased volumes in RTN4RL2^{-/-} IHCs (RTN4RL2^{+/+}: $0.36 \pm 0.01 \mu\text{m}^3$, SD = $0.16 \mu\text{m}^3$, n = 160, N = 2 vs RTN4RL2^{-/-}: $0.26 \pm 0.01 \mu\text{m}^3$, SD = $0.19 \mu\text{m}^3$, n = 249, N = 3; p < 0.001, Mann-Whitney-Wilcoxon test). (F) Percentage of Ctbp2 puncta juxtaposing Homer1 is slightly decreased in RTN4RL2^{-/-} mice (RTN4RL2^{+/+}: $96.7 \pm 1.45 \%$, SD = 5.44 %, n = 14, N = 2 vs RTN4RL2^{-/-}: $89.5 \pm 2.05 \%$, SD = 9.18 %, n = 20, N = 3; p = 0.03, Mann-Whitney-Wilcoxon test). Data is presented as mean ± SEM. Box-whisker plots show the median, 25/75 percentiles (box), and 10/90 percentiles (whiskers). Individual data points are overlaid.

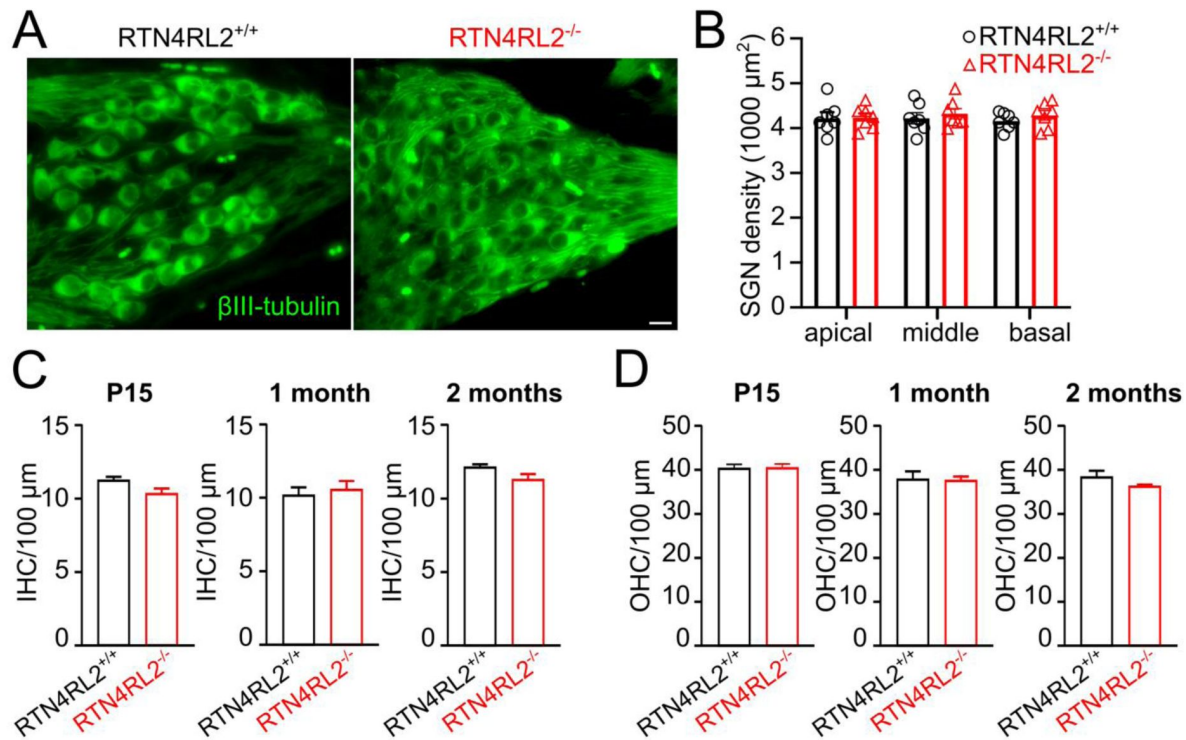


Figure 2-figure supplement 1.

Cochlear cell densities are not changed in RTN4RL2^{-/-} mice.

(A) Mid/Basal-modiolar sections labeled for βIII-tubulin (green, neurons) from p40 RTN4RL2^{+/+} and RTN4RL2^{-/-} mice. Exemplary section for RTN4RL2^{-/-} is the zoomed out image presented in [figure 1C](#). (B) Quantitative analysis shows that the density of SGN cell bodies are similar between RTN4RL2^{-/-} and control cochleae. Data is presented as mean ± SD; N = 7 per group. Scale bar = 10 μm. (C, D) IHC (C) and OHC (D) densities are not affected in P15, 1 month and 2 months old RTN4RL2^{-/-} mice.

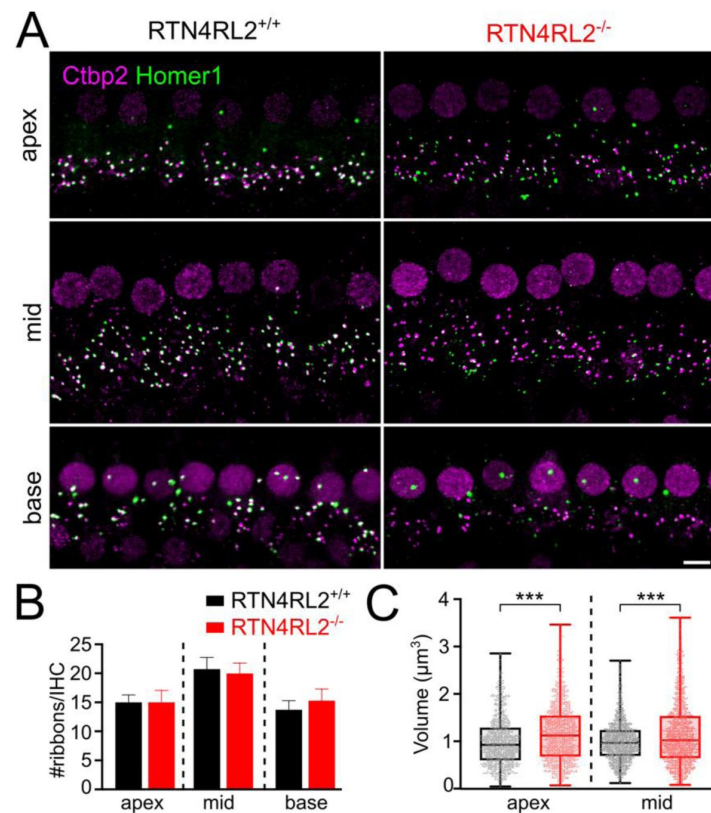


Figure 2-figure supplement 2.

Intact number but enlarged size of the ribbons in RTN4RL2^{-/-} IHCs.

(A) Maximum intensity projections of representative IHCs from apical, mid and basal regions of RTN4RL2^{+/+} (left) and RTN4RL2^{-/-} (right) cochleae of p21-30 mice. Synapses are visualized by staining against Ctip2/Ribeye (ribbons) and Homer1 (PSDs). Scale bar = 5 μm. (B) The number of the ribbons is not affected along the tonotopic axis in RTN4RL2^{-/-} cochleae. (C) The size of the ribbons is increased in IHCs of both apical and middle turns in RTN4RL2^{-/-} cochleae ($p < 0.001$, Mann-Whitney-Wilcoxon test). $N = 6$ animals/genotype. Box-whisker plots show the median, 25/75 percentiles (box) and the range (whiskers). Individual data points are overlaid.

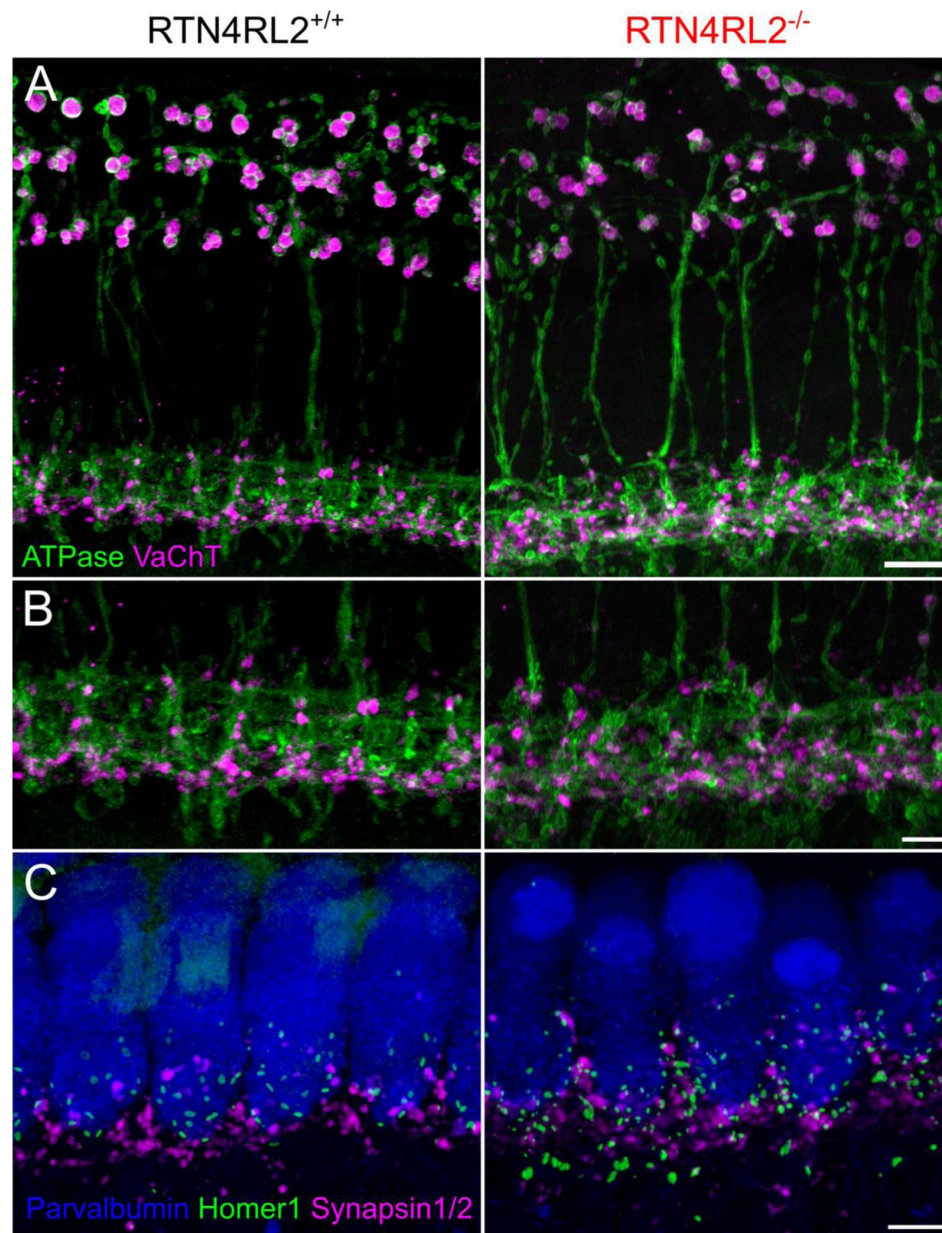


Figure 2-figure supplement 3.

Efferent innervation pattern in *RTN4RL2*^{-/-} cochleae.

(A) Maximum intensity projections of confocal stacks of outer and inner hair cell regions at the apical region of the cochlea from p21-30 *RTN4RL2*^{+/+} (left) and *RTN4RL2*^{-/-} (right) mice. SGN fibers/terminals and efferent terminals are visualized staining for Na⁺/K⁺ ATPase and vesicular acetylcholine transporter (VaChT), respectively. Scale Bar = 10 μ m (B) Same as (A) but zoomed into the IHC region. Scale Bar = 5 μ m (C) Maximum intensity projections of confocal stacks of a row of apical IHC region from p21 *RTN4RL2*^{+/+} (left) and *RTN4RL2*^{-/-} (right) mice. Presynaptic efferent terminals were stained with an anti-synapsin1/2 antibody. Scale bar = 5 μ m.

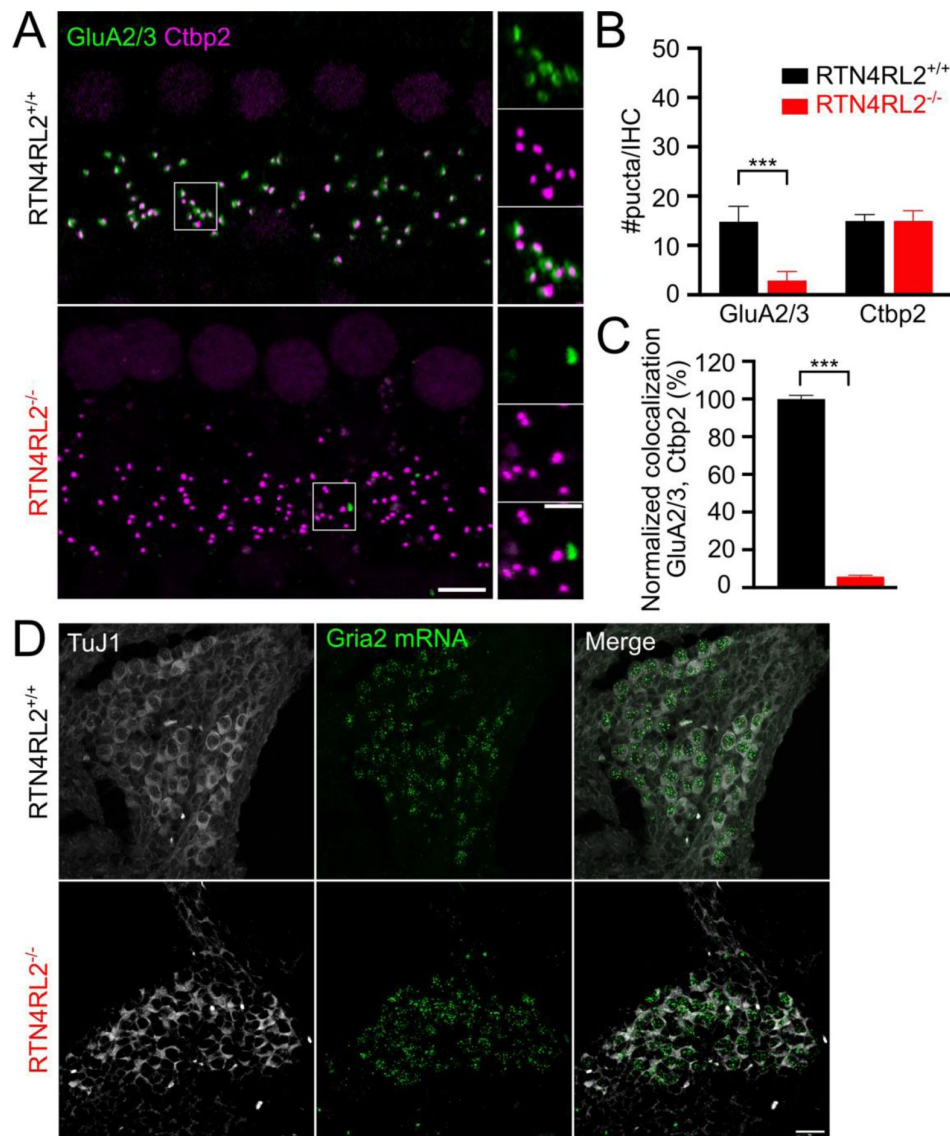


Figure 3.

Reduced GluA2/3 signal juxtaposing presynaptic ribbons in IHCs of RTN4RL2^{-/-} mice.

(A) Maximum intensity projections of representative IHC regions from 1- to 1.5-month- old RTN4RL2^{+/+} (top) and RTN4RL2^{-/-} (bottom) mouse cochleae immunolabeled against Ctb2/Ribeye (ribbons) and GluA2/3 (AMPA receptors of PSD). Scale bar = 5 μ m. The zoom-in regions marked with the white rectangles are presented on the right-hand side. Scale bar = 2 μ m. (B) The number of the GluA2/3 positive puncta is drastically reduced in RTN4RL2^{-/-} mice despite the maintained number of presynaptic ribbons ($p < 0.001$, Mann-Whitney-Wilcoxon test). (C) Disrupted colocalization of Ctb2/Ribeye and GluA2 immunofluorescence puncta at IHCs of RTN4RL2^{-/-} mice ($p < 0.001$, Mann-Whitney-Wilcoxon test). N = 6 animals/genotype. (D) Representative images of RNAscope ISH from p4 mice show maintained expression of Gria2 (red dots) in the SGN somata of RTN4RL2^{-/-} mice. Scale bar = 20 μ m.

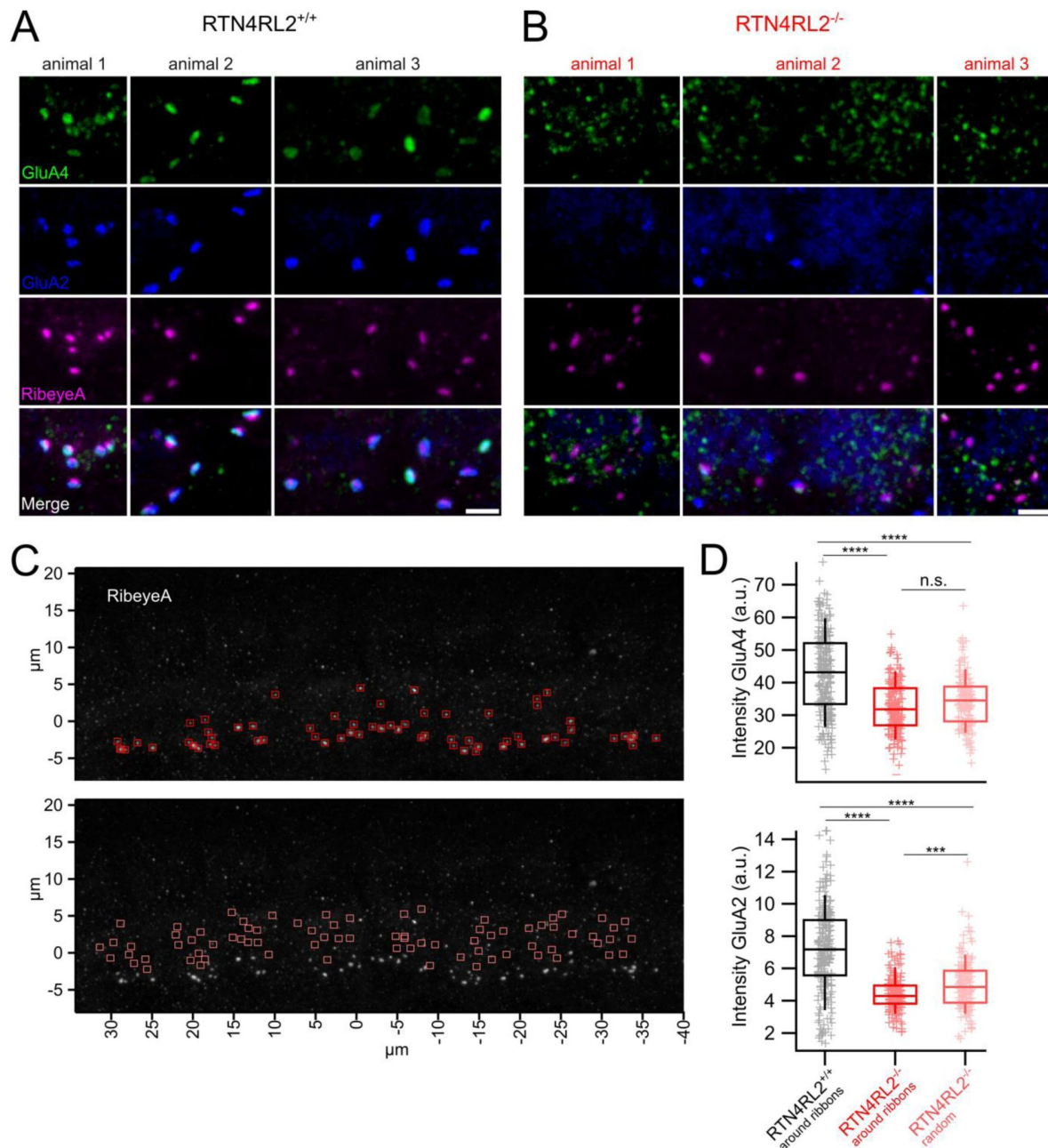


Figure 3-figure supplement 1.

No apparent GluA4 signal juxtaposing presynaptic ribbons at IHC synapses of RTN4RL2^{-/-} mice.

(**A, B**) Maximum intensity projections of a few imaging planes from apical IHC synaptic regions of 14-week-old N = 3 RTN4RL2^{+/+} (**A**) and N = 3 RTN4RL2^{-/-} (**B**) mice immunostained against GluA4, GluA2 and RibeyeA. Scale bars = 2 μ m. (**C**) Fluorescence intensities of GluA4 and GluA2 channels were analyzed as the sum of all pixel intensities in approximately 1 $\times$ 1 $\times$ 1 μ m³ volumes around the center of mass of RibeyeA immunofluorescent puncta (red boxes, top image). Background GluA4 and GluA2 fluorescence signals were calculated in the same way but outside the ribbons, at random spots inside IHCs (pink boxes, bottom image). Images show the maximum intensity projection of the IHC region in an RTN4RL2^{-/-} mouse. (**D**) Fluorescence intensities of GluA4 (top) and GluA2 (bottom) channels at ribbons of RTN4RL2^{-/-} mice do not show a significant increase compared to the background (RTN4RL2^{-/-} random) but are significantly decreased compared to RTN4RL2^{+/+} (Mann-Whitney-Wilcoxon test followed by Bonferroni multiple-comparison correction). Box-whisker plots show the median, 25/75 percentiles (box) and 10/90 percentiles (whiskers). Individual data points are overlaid.

et al., 2021 [↗](#); Lu et al., 2024 [↗](#)), the spatial resolution of SBEM allows reliable ribbon synapse identification and neurite reconstruction (**Figure 4B** [↗](#) and C). In these three datasets, we have traced all neural fibers from the habenula perforata (three in each dataset). This resulted in 133 fibers, of which 115 were classified as peripheral neurites of type I SGN based on their radial calibers and extensive myelination after entering the habenula perforata (**Figure 4D** [↗](#)). We observed a substantial number of neurites that did not engage the IHCs (non-synaptic neurites) in the RTN4RL2^{-/-} organs of Corti (23 out of 83) in contrast to the wild-type control (3 out of 32, **Figure 4E** [↗](#)). Although most non-synaptic neurites were found in the inner spiral bundle, they failed to reach IHC basal lateral poles to form contacts (17 out of 23 in RTN4RL2^{-/-} animals). This result might provide a plausible explanation for the “orphan” Homer1 puncta shown by immunohistochemistry in RTN4RL2^{-/-} organs of Corti. Note that the presence of non-synaptic neurites in the RTN4RL2^{-/-} animal was not associated with a profound reduction in ribbon synapses (**Figure 2C** [↗](#), **Figure 2-figure supplement 2B** [↗](#), **Figure 4-figure supplement 1A** [↗](#)), suggesting a different scenario than deafferentation due to excitotoxic synaptopathy as recently reported (Moverman et al., 2023 [↗](#)). Nevertheless, nearly all traced radial fibers were found predominantly unbranched in both the RTN4RL2^{-/-} (97%) and the wild-type control mice (94%; **Figure 4F** [↗](#)).

Deletion of RTN4RL2 results in depolarized shift of the Ca²⁺ channel activation in IHCs

Next, we tested for potential effects of RTN4RL2 deletion on the presynaptic IHC function. We performed whole-cell patch-clamp recordings from IHCs of p21-29 RTN4RL2^{+/+} and RTN4RL2^{-/-} mice. First, we recorded voltage-gated Ca²⁺ currents by applying step depolarizations with 5 mV increment in ruptured patch-clamp configuration (**Figure 5A** [↗](#), see Materials and Methods). We did not observe any change in the maximal Ca²⁺ current amplitude (**Figure 5B** [↗](#), C), yet noticed small (~ +3 mV) but consistent and statistically significant depolarized shift of the voltage of half-maximal Ca²⁺ channel activation (V_{half} ; **Figure 5D** [↗](#), E). The voltage sensitivity of the channels was not changed in RTN4RL2^{-/-} IHCs (**Figure 5F** [↗](#)). Next, we probed Ca²⁺ influx triggered exocytosis from IHCs by applying depolarizations of varying durations to voltages saturating Ca²⁺ influx in IHCs of both genotypes (-17 mV) and recording exocytic membrane capacitance changes (ΔC_m) in the perforated patch-clamp configuration (**Figure 5G** [↗](#)). Both fast (up to 20 ms depolarization) and sustained components of exocytosis remained intact in the mutant, indicating unaffected readily releasable pool and replenishment of the vesicles in RTN4RL2^{-/-} IHCs (**Figure 5H** [↗](#)).

We then turned to study single AZ function by combining whole cell ruptured patch-clamp with spinning disc confocal imaging of presynaptic Ca²⁺ influx, as described previously (Ohn et al., 2016 [↗](#)). TAMRA conjugated Ctb2 binding dimeric peptide and the low affinity Ca²⁺ indicator Fluo4-FF (K_D : 10 μ M) were loaded into the cell via the patch pipette and Ca²⁺ influx at single AZs was visualized in the form of hotspots by applying voltage ramp depolarizations to the IHCs and simultaneously imaging Fluo4-FF fluorescence near labeled ribbons (**Figure 6A** [↗](#), see Materials and methods). Maximal Ca²⁺ influx at single AZs tended to be higher in IHCs of RTN4RL2^{-/-} mice but the difference did not reach statistical significance (**Figure 6B** [↗](#), Bⁱ). The V_{half} of Ca²⁺ channel clusters at single AZs showed depolarized shift (~ +5 mV), which was statistically significant despite the large variability of V_{half} across different AZs (**Figure 6C** [↗](#), Cⁱ). Furthermore, we did not observe any mismatch in Ctb2 positive puncta and evoked Ca²⁺ hotspots, indicating the correct localization of Ca²⁺ channels at the AZs. We verified this by performing immunostainings of Ca_v1.3, Ctb2 and Bassoon and did not detect any apparent misalignment of these protein clusters (**Figure 6D** [↗](#)). In summary, deletion of RTN4RL2 causes a depolarized shift of the activation of presynaptic Ca²⁺ influx in IHCs as demonstrated at the single AZ and whole cell

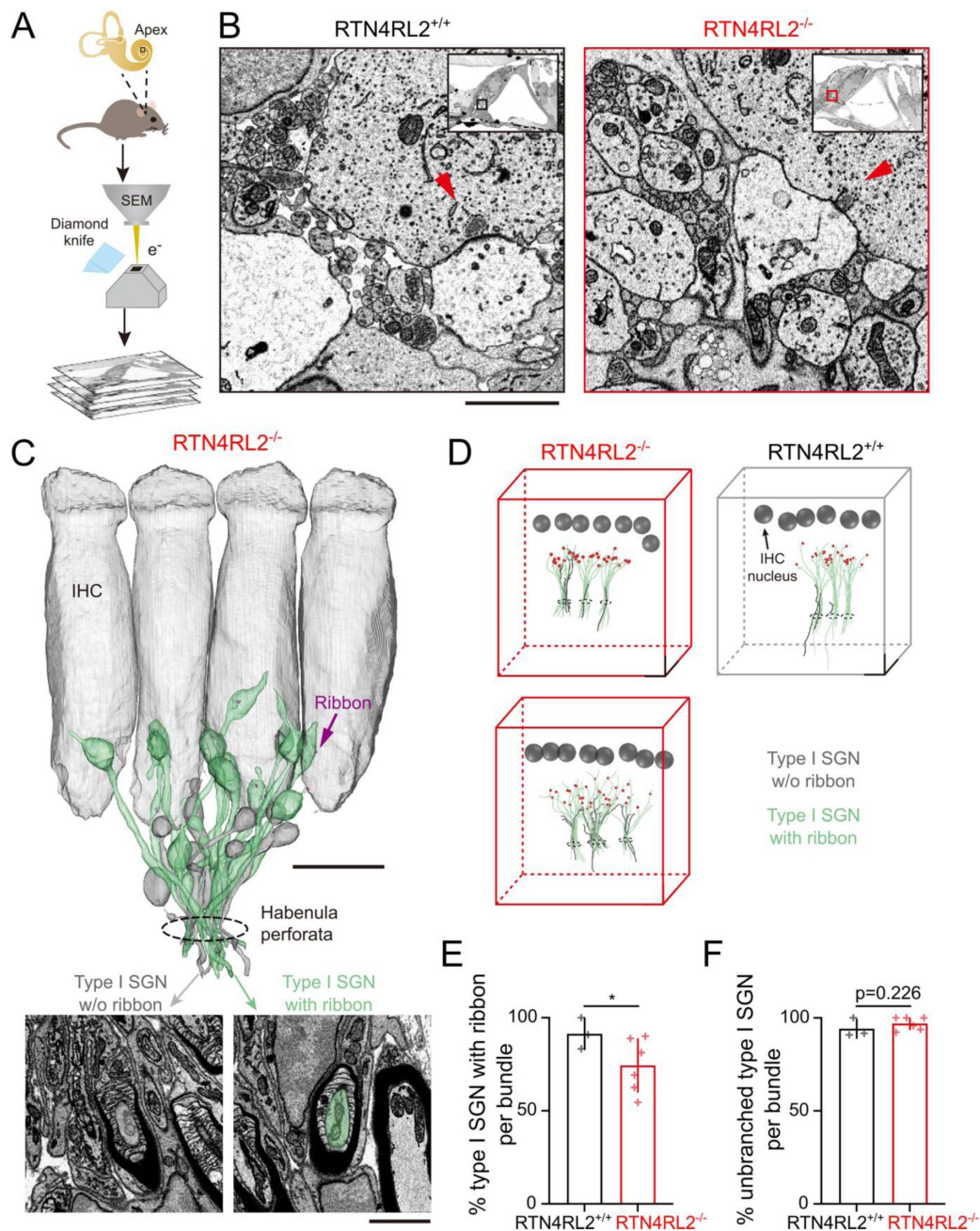


Figure 4.

Additional non-synaptically engaged SGN neurites in the cochlea of RTN4RL2^{-/-} mice.

(A) Workflow of SBEM imaging at the mouse apical cochlear region. (B) Example images of neurites beneath IHCs from the RTN4RL2^{+/+} (left) and RTN4RL2^{-/-} (right) mice. Synaptic ribbons are indicated with red arrows. The regions of interest were magnified from single sections of SBEM datasets (insets). Scale bar = 2 μ m. (C) 3D rendering of afferent fiber reconstruction with ribbons (red), showing both synaptic (green, type I SGN with ribbon) and non-synaptic (grey, type I SGN without ribbon) populations in the RTN4RL2^{-/-} mouse. Scale bar = 10 μ m. Inset: EM images show the SGN segments on which myelin sheaths start to form. Scale bar = 2 μ m. (D) Display of classified radial fibers in the RTN4RL2^{+/+} (right) and RTN4RL2^{-/-} (left and bottom) animals. All fibers were traced from the habenula perforata (cycles) before classification to avoid bias to terminal types. Scale bar = 10 μ m. (E) Percentage of radial fibers with ribbon per bundle (RTN4RL2^{+/+}: 91.41 $\pm$ 8.35 %, n = 3 bundles, N = 1 vs RTN4RL2^{-/-}: 74.40 $\pm$ 14.58%, n = 6 bundles, N = 2; p = 0.032, unpaired t-test). (F) Percentage of unbranched radial fibers per bundle (RTN4RL2^{+/+}: 94.19 $\pm$ 5.04 %, n = 3 bundles, N = 1 vs RTN4RL2^{-/-}: 96.98 $\pm$ 3.49 %, n = 6 bundles, N = 2; p = 0.226, unpaired t-test).

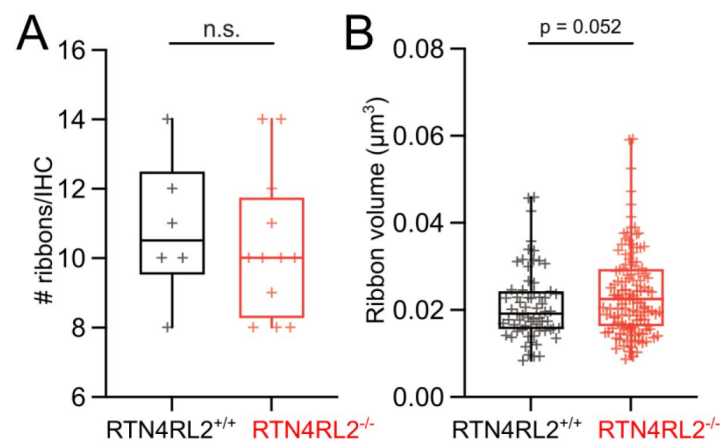


Figure 4-figure supplement 1.

Quantification of ribbon number and volume in SBEM reconstructions.

(A) The number of ribbons is not changed in RTN4RL2^{-/-} IHCs (RTN4RL2^{-/-}: 10.3 ± 0.61 , SD = 2.1, n = 12, N = 2 vs. RTN4RL2^{+/+}: 10.8 ± 0.83 , SD = 2.04, n = 6, N = 1; p = 0.54, Mann-Whitney-Wilcoxon test). (B) Ribbon volumes tend to be larger in RTN4RL2^{-/-} IHCs without reaching statistical significance (RTN4RL2^{-/-}: $0.024 \pm 0.001 \mu\text{m}^3$, SD = $0.01 \mu\text{m}^3$, n = 125 ribbons in 12 IHCs, N = 2 vs. RTN4RL2^{+/+}: $0.021 \pm 0.001 \mu\text{m}^3$, SD = $0.008 \mu\text{m}^3$, n = 65 ribbons in 6 IHCs, N = 1; p = 0.06, Mann-Whitney-Wilcoxon test). Box-whisker plots show the median, 25/75 percentiles (box) and the range (whiskers). Individual data points are overlaid.

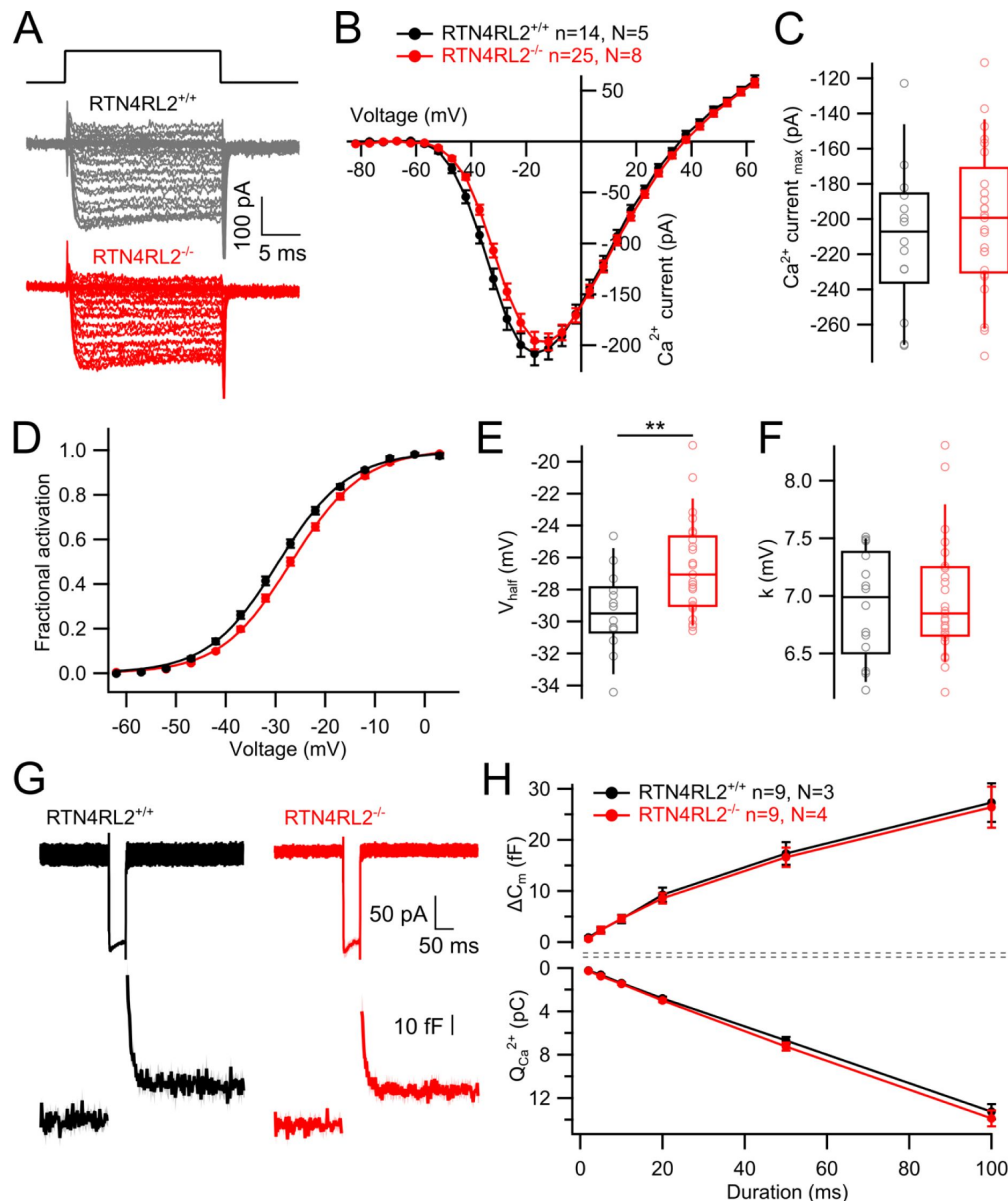


Figure 5.

Shifted operation range of Ca^{2+} channels but intact exocytosis in IHCs of *RTN4RL2*^{-/-} mice.

(A) Representative current traces from IHCs of *RTN4RL2*^{+/+} (top, black) and *RTN4RL2*^{-/-} (bottom, red) evoked by step depolarizations. (B) Average Ca^{2+} current-voltage relationships (IV curves) in *RTN4RL2*^{+/+} and *RTN4RL2*^{-/-} IHCs. (C) Maximal Ca^{2+} current amplitude is not changed in *RTN4RL2*^{-/-} IHCs (*RTN4RL2*^{+/+}: 210 ± 10.9 pA, SD = 40.9 pA, n = 14, N = 5 vs *RTN4RL2*^{-/-}: -200 ± 8.19 pA, SD = 40.9 pA, n = 25, N = 8; p = 0.47, Student's t-test). (D) Fractional activation curves of Ca^{2+} channels calculated from IVs show depolarized shift in channel activation in *RTN4RL2*^{-/-} IHCs. (E) Voltage of half maximal activation obtained from Boltzmann fit of the curves from (D) is more positive in *RTN4RL2*^{-/-} IHCs (*RTN4RL2*^{+/+}: -29.4 ± 0.66 mV, SD = 2.48 mV, n = 14, N = 5 vs *RTN4RL2*^{-/-}: -26.6 ± 0.59 mV, SD = 2.94 mV, n = 25, N = 8; p = 0.004, Student's t-test). (F) Voltage sensitivity (k) is not changed in *RTN4RL2*^{-/-} IHCs (*RTN4RL2*^{+/+}: 6.92 ± 0.12 mV, SD = 0.46 mV, n = 14, N = 5 vs *RTN4RL2*^{-/-}: 6.98 ± 0.1 mV, SD = 0.51 mV, n = 25, N = 8; p = 0.93, Mann-Whitney-Wilcoxon test). (G) Average current traces evoked by 50 ms depolarization to -17 mV (top row) and resulting capacitance response (bottom row) from *RTN4RL2*^{+/+} (left, black) and *RTN4RL2*^{-/-} (right, red) IHCs. Shaded areas represent $\pm$ SEM. (H) Exocytic capacitance change (ΔC_m , top) and corresponding Ca^{2+} charge ($Q_{\text{Ca}^{2+}}$, bottom) evoked by depolarizations (to -17 mV) of various durations (2, 5, 10, 20, 50, 100 ms). Box-whisker plots show the median, 25/75 percentiles (box) and 10/90 percentiles (whiskers). Individual data points are overlaid.

levels. This is expected to elevate the sound pressure levels required to reach a comparable activity of synaptic transmission for sound encoding and predicts elevated auditory thresholds which we tested by recordings of auditory brainstem responses.

RTN4RL2 is essential for normal hearing

In order to evaluate auditory systems function in RTN4RL2^{-/-} mice, we recorded auditory brainstem responses (ABRs) at the age of 2-4 months. ABRs were elicited in response to 4, 8, 16, 32 kHz tone bursts and click stimuli of increasing sound pressure levels. We determined the sound thresholds by identifying the lowest sound pressure level which resulted in detectable ABR waveform. We found ABR thresholds to be significantly increased by around 30-45 dB across all the tested frequencies and in response to the click stimulations in RTN4RL2^{-/-} mice compared to the wild-type controls (**Figure 7**). RTN4RL2^{+/-} mice showed an intermediate phenotype between the RTN4RL2^{-/-} and RTN4RL2^{+/+} mice with a lower yet significant increase of ABR thresholds of approximately 10-15 dB at 4, 16 and 32 kHz frequencies.

Discussion

In this study, we investigated the role of the Nogo/RTN4 receptor homolog RTN4RL2 in the cochlea. Consistent with the previous reports, we detected RTN4RL2 expression in IHCs and SGNs. Upon RTN4RL2 deletion, we observed alterations of the IHC-SGN synapses (**Figure 8**). Presynaptically, Ca²⁺ channel activation was shifted towards depolarized voltages and ribbon size was enlarged. IHC exocytosis was unaltered when probed at saturating depolarizations. At the postsynaptic side, PSDs juxtaposing presynaptic ribbons were smaller compared to the controls and seemed to be deficient of GluA2-4, despite the expression of Gria2 mRNA in RTN4RL2^{-/-} SGNs. Additionally, we observed PSDs which resided further from the IHC membrane and did not juxtapose the presynaptic AZs in RTN4RL2^{-/-} mice. These PSDs potentially belong to the type I SGN neurites that ended in the inner spiral bundle without contacting the IHCs, as observed in SBEM reconstructions of RTN4RL2^{-/-} SGNs. Finally, ABR thresholds were elevated in RTN4RL2^{-/-} mice indicating that RTN4RL2 is required for normal hearing.

RTN4 receptors have been implicated as presynaptic adhesion molecules based on their transsynaptic interactions with postsynaptically enriched brain-specific angiogenesis inhibitor (BAI) adhesion GPCRs (Stephenson et al., 2013; Wang et al., 2021). Despite this, at least RTN4R has been proposed to function postsynaptically, given that its knockdown leads primarily to postsynaptic changes (Wills et al., 2012). The interpretation of RTN4R function and localization is further complicated by their enrichment in both pre- and postsynaptic terminals (Wang et al., 2002b). Similarly, we observed RTN4RL2 expression in both IHCs and SGNs. Interestingly, a recent study described a hearing impairment in BAI1 deficient mice and suggested that postsynaptically functioning BAI1 is important for correct localization of AMPA receptor subunits in spiral ganglion neurons based on the absence or decrease of different AMPA receptor subunits at the SGN postsynapses in BAI1 deficient mice (Carlton et al., 2024). Similarly, our data from RTN4RL2^{-/-} mice show the absence of GluA2-4 positive puncta juxtaposing presynaptic AZs. Importantly, the BAI1 knock-out model in the study by Carlton et al. eliminated specifically long isoform of BAI1, which contains extracellular thrombospondin repeats (TSRs) important for mediating transsynaptic interactions, while leaving the short isoform - containing the transmembrane repeats and intracellular domain of the receptor - intact (Wang et al., 2021; Carlton et al., 2024). Along with the evidence for transsynaptic interaction between RTN4Rs and BAIs (Wang et al., 2021), this raises the possibility that the GluA2-4 AMPA receptor subunit localization to SGN PSDs requires the interaction of presynaptic RTN4RL2 and postsynaptic BAI1. Likewise, the changes of presynaptic Ca²⁺ channel properties and ribbons in IHCs could result from direct presynaptic action of RTN4RL2. Alternatively, RTN4RL2 could act postsynaptically in

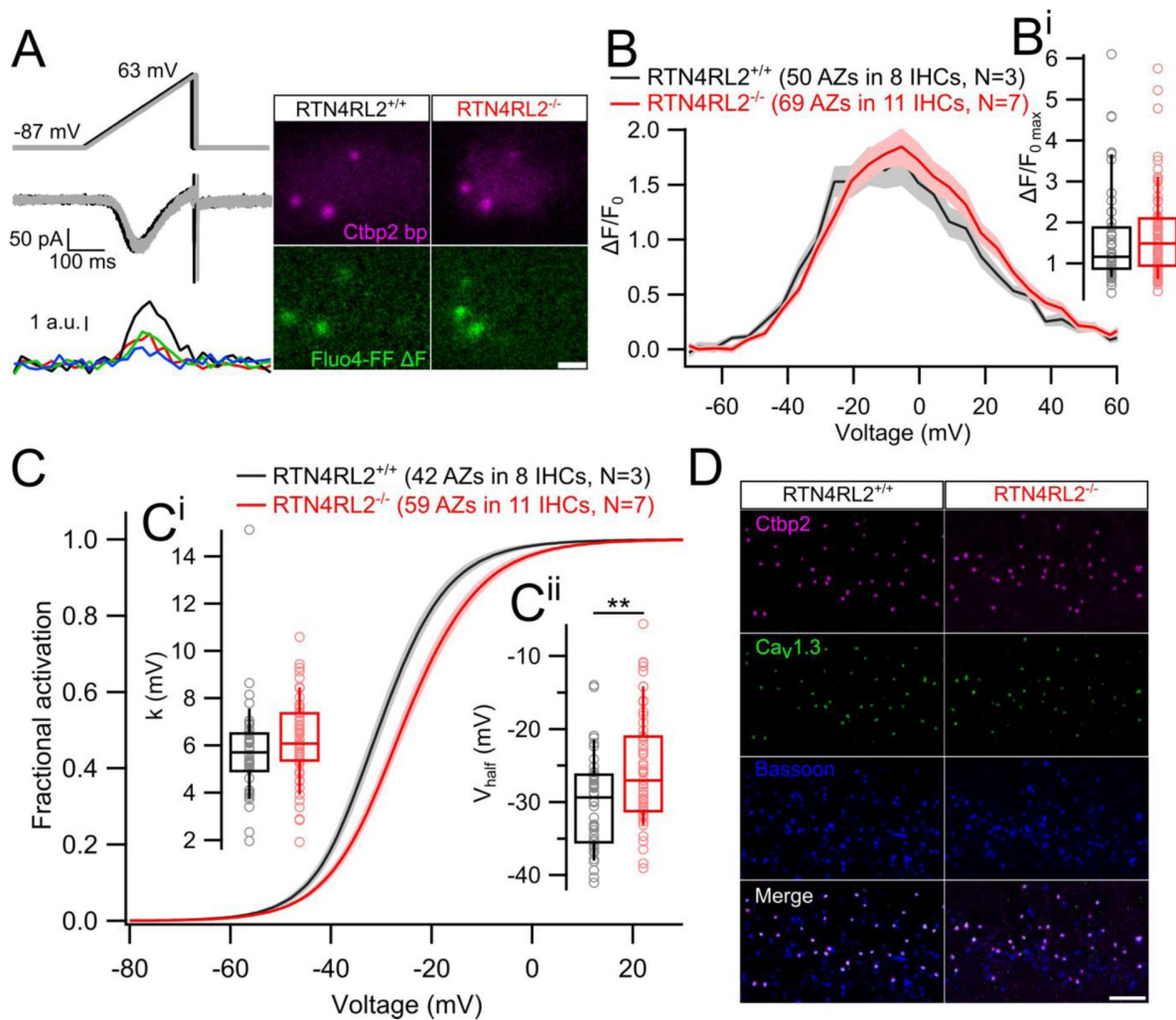


Figure 6.

Depolarized shift of Ca²⁺ channel activation at single AZs but intact presynaptic organization in RTN4RL2^{-/-} IHCs.

(A) Voltage ramp stimulation protocols (top), evoked whole cell currents (middle) and the presynaptic hotspots of Fluo4-FF fluorescence (bottom) of a representative IHC recording. Black and grey colors represent the two stimulations, one being 5 ms shifted over the other. Images on the right show single imaging planes of representative RTN4RL2^{+/+} (left) and RTN4RL2^{-/-} (right) IHCs filled with TAMRA conjugated Ctbp2 binding peptide (Ctbp2 bp) and Fluo4-FF Ca²⁺ dye. Ca²⁺ hotspots are visualized by subtracting the average of baseline planes from the average of 5 planes during stimulation. Scale bar = 2 μ m.

(B) Average fluorescence-voltage relationships of Ca²⁺ influx at single AZ from RTN4RL2^{+/+} and RTN4RL2^{-/-} IHCs show no difference in the maximal Ca²⁺ amplitude (**Bⁱ**; RTN4RL2^{+/+}: 1.6 ± 0.17 , SD = 1.2, n = 50 AZs vs RTN4RL2^{-/-}: 1.7 ± 0.13 pA, SD = 1.09, n = 69 AZs; p = 0.24, Mann-Whitney-Wilcoxon test). Shaded areas represent $\pm$ SEM.

(C) Average fractional activation curves of Ca²⁺ channels at single AZs show intact voltage sensitivity (**Cⁱ**; RTN4RL2^{+/+}: 5.79 ± 0.32 mV, SD = 2.04 mV, n = 42 AZs vs RTN4RL2^{-/-}: 6.25 ± 0.22 mV, SD = 1.7 mV, n = 59 AZs; p = 0.06, Mann-Whitney-Wilcoxon test) but depolarized shift of V_{half} (**Cⁱⁱ**; RTN4RL2^{+/+}: -30 ± 1 mV, SD = 6.5 mV, n = 42 AZs vs RTN4RL2^{-/-}: -25.5 ± 0.98 mV, SD = 7.49 mV, n = 59 AZs; p = 0.002, Student's t-test) in RTN4RL2^{-/-} IHCs. Shaded areas represent $\pm$ SEM.

(D) Representative immunolabelings of presynaptic proteins show no apparent mislocalization in RTN4RL2^{-/-} IHCs. Scale bar = 5 μ m. Box-whisker plots show the median, 25/75 percentiles (box) and 10/90 percentiles (whiskers). Individual data points are overlaid.

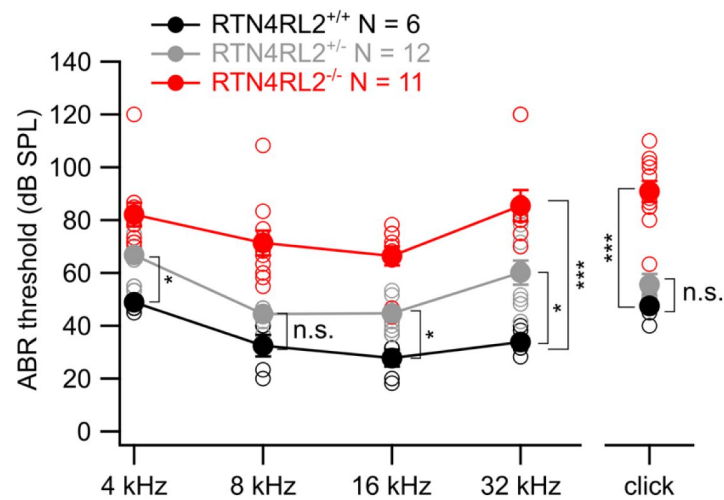


Figure 7.

Elevated acoustic thresholds in RTN4RL2^{-/-} mice.

ABR thresholds were measured in response to 4, 8, 16, 32 kHz tone bursts and click stimuli. ABR thresholds of individual animals are shown in open circles on top of the mean ± SEM. Statistical significances are reported as *p < 0.05, ***p < 0.001, Kruskal-Wallis followed by Dunn's multiple comparison test.

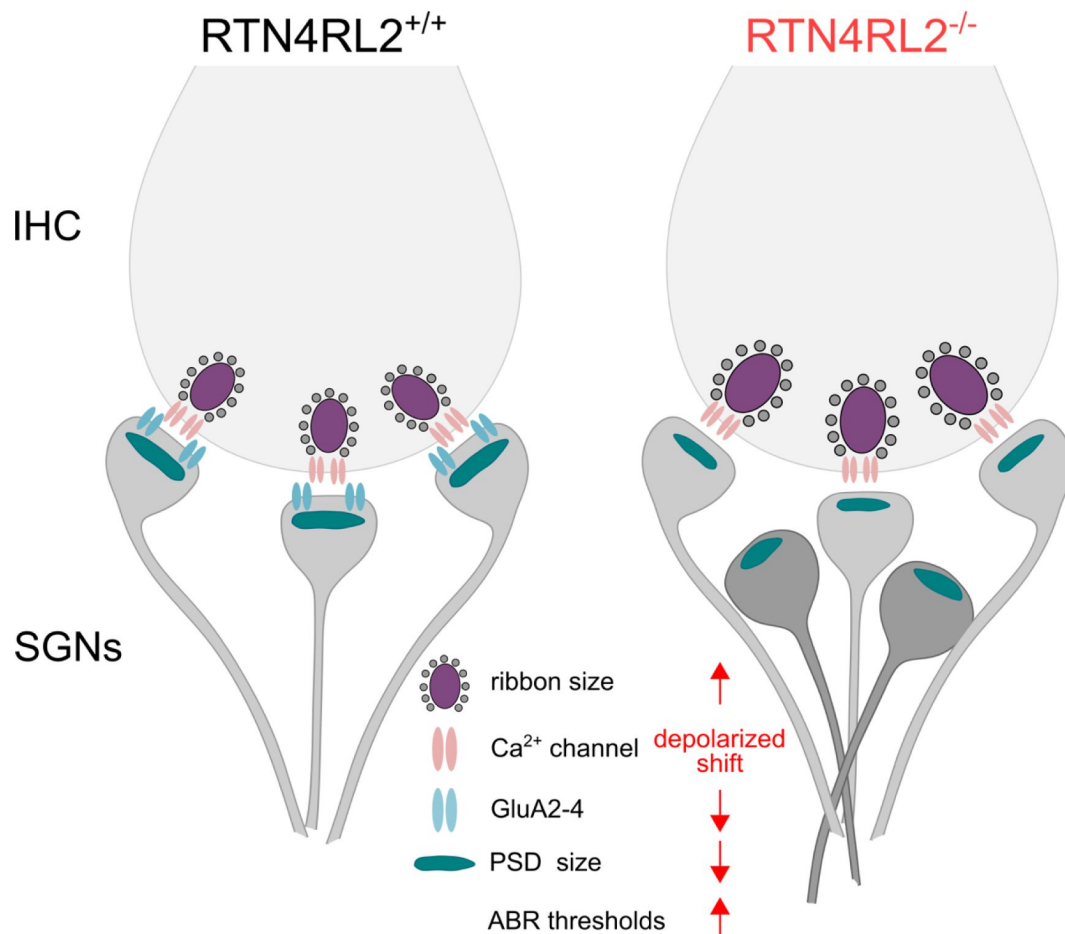


Figure 8.

Schematic illustration of the key structural and functional changes in the auditory periphery of RTN4RL2^{-/-} mice.

RTN4RL2^{-/-} mice display enlarged synaptic ribbons and depolarized shift in the activation of presynaptic Ca²⁺ channels in IHCs, as well as reduced size of PSDs juxtaposing presynaptic ribbons. RTN4RL2 deficiency further leads to a decrease in GluA2-4 AMPA receptor subunits at PSDs and results in a subset of type I SGN neurites that reach the inner spiral bundle but do not engage the IHCs.

SGNs with an impact on transsynaptic signaling to IHCs. Presynaptic changes upon postsynaptic manipulation have been reported previously. For instance, the deletion of postsynaptically expressed Pou4f1 transcription factor resulted in hyperpolarized shift of IHC Ca^{2+} channels, as well as in changes of the presynaptic heterogeneity of IHC AZs (Sherrill et al., 2019 [DOI](#)). Similarly, presynaptic ribbon sizes were altered in IHCs of mice, which exhibited conditional deletion of Runx1 transcription factor from SGNs (Shrestha et al., 2023 [DOI](#)). Further experiments, employing mouse mutants with specific RTN4RL2 deletion in IHCs or SGNs will allow to evaluate the precise presynaptic and postsynaptic roles of RTN4RL2 including a possible transsynaptic signaling to IHC AZs.

We propose that the afferent synaptic changes contribute to the severe hearing impairment in RTN4RL2^{-/-} mice. The phenotype of increased ABR thresholds could partially be explained by the depolarized activation of Ca^{2+} channels in IHCs. As it has been shown, the apparent Ca^{2+} dependence of glutamate release from IHCs on average follows a near linear relationship (Wong et al., 2014 [DOI](#); Özçete and Moser, 2021 [DOI](#); Jaime Tobón and Moser, 2023b, 2023a). Therefore, due to the depolarized shift of the Ca^{2+} channel activation, the glutamate release from IHCs would require stronger stimuli. While exocytosis, probed by depolarizing the IHCs to -17 mV and recording cell membrane capacitance changes, remained intact in RTN4RL2^{-/-} mice, it could still be affected at milder depolarizations (e.g., -50 mV to -40 mV), which correspond to the receptor potential range of the IHCs. This could result in increased sound thresholds, as shown in Ribeye knock-out mice, where depolarized activation of Ca^{2+} channels has been associated with higher firing thresholds and lower spontaneous rates in SGNs, although a less severe hearing phenotype was observed in those mice (Jean et al., 2018 [DOI](#)). Furthermore, we have shown recently that a direct manipulation of the voltage dependence of $\text{Ca}_v1.3$ Ca^{2+} channel activation results in changes of SGN firing and potentially auditory thresholds, further supporting our hypothesis of sound threshold elevation in RTN4RL2^{-/-} mice being caused by the depolarized activation of Ca^{2+} channels in IHCs (Karagulyan et al., 2025 [DOI](#)).

In addition or alternatively, the increase in auditory thresholds could be caused by the deficiency of GluA2-4 AMPA receptors of the SGN postsynapses in RTN4RL2^{-/-} mice. Compared to GluA3 KO mice, which display reduced GluA2 signal at SGN PSDs but intact auditory thresholds, the increased auditory thresholds in RTN4RL2^{-/-} mice could additionally be caused by the lack of GluA4 subunits juxtaposing presynaptic ribbons, which, in contrast, were shown to be upregulated in GluA3 KO mice (Rutherford et al., 2023 [DOI](#)). Given the observed synaptic changes, it seems less likely that the hearing impairment would have resulted from the intrinsic changes of SGN firing. Further experiments, where SGNs would be stimulated directly via *ex vivo* patch-clamp or optogenetically could help to probe for functional deficits in SGNs of RTN4RL2^{-/-} mice. Another interesting observation was the additional PSDs not juxtaposing presynaptic ribbons that were detected around IHCs of RTN4RL2^{-/-} mice. These “orphan” PSDs seem not to be part of the efferent synapses formed between LOC/MOCs and SGNs as we verified by simultaneous anti-Homer1 and anti-synapsin immunolabelings. To check the SGN connectivity we performed SBEM in RTN4RL2^{+/+} and RTN4RL2^{-/-} cochlear samples and observed a high percentage of non-synaptic SGN neurites in the reconstructed neurons of inner spiral bundles in RTN4RL2^{-/-} samples. A sizable subset of the non-synaptic SGN boutons did not make a contact with IHCs, potentially housing the “orphan” PSDs observed around the IHCs in our immunolabeled samples. How exactly those non-synaptic neurites appear remains to be investigated. SGN neurite retraction from IHCs has been demonstrated in noise-exposed animals, followed by the degeneration of the SGN somata and axons (Kujawa and Liberman, 2009 [DOI](#), 2015 [DOI](#); Moverman et al., 2023 [DOI](#)). This process however results in up to 50% degeneration of the ribbon synapses. Interestingly, in RTN4RL2^{-/-} mice the number of the ribbon synapses as well as the density of the SGN somata was largely intact. This indicates that the non-synaptic neurites do not result from synaptic loss as found in hereditary or acquired synaptopathy (Roux et al., 2006 [DOI](#); Ruel et al., 2008 [DOI](#); Seal et al., 2008 [DOI](#); Kujawa and Liberman, 2009 [DOI](#)). Although it is plausible that the non-synaptic SGN neurites may fail to reach the IHCs during initial development, it is unlikely that this would selectively impact only a subset

of neurons and still allow the Homer1-positive patches to develop in the absence of presynaptic components. Alternatively, it is known that major synaptic pruning takes place in developing SGNs, whereby up to 50% of the synapses are lost between the IHCs and the SGNs (Defourny et al., 2013 [↗](#); Coate et al., 2019 [↗](#)), suggesting that the non-synaptic SGN neurites of RTN4RL2^{-/-} mice could be a result of failed pruning of SGNs during the development. Yet how this would then comply with normal synapse and SGN counts remains to be elucidated. Future studies will be required to understand the fate of type I SGNs in RTN4RL2^{-/-} mice during development by mapping their full-length periphery projections with large scale imaging tools.

Future work will be required to identify the RTN4RL2 ligands in the cochlea. To this date, the interaction partners of RTN4Rs are not very well understood. While RTN4R in-trans interactions include NogoA, MAG, OMgp, chondroitin sulfate proteoglycans, the only well established ligand for RTN4RL2 remains MAG (Fournier et al., 2001 [↗](#); Domeniconi et al., 2002 [↗](#); Wang et al., 2002a [↗](#); Venkatesh et al., 2005 [↗](#); Schwab, 2010 [↗](#)). Other attractive candidate ligands for RTN4Rs are the BAIs. BAI-RTN4R interaction has been suggested to mediate dendritic arborization, axonal elongation, synapse formation in IPSCs (Wang et al., 2021 [↗](#)) and, in the case of this study GluA2-4 AMPA receptor subunit localization to the SGN postsynaptic densities. Finally, RTN4RL2 has been proposed to interact with chondroitin-sulfate proteoglycan Versican to control the amount of skin innervation by DRG neurites (Bäumer et al., 2014 [↗](#)). While to our knowledge, attempts have yet to be made for detecting Versican signal around the IHCs, other extracellular matrix proteins such as aggrecan, brevican, Tenascin-R, Tenascin-C, hyaluron and proteoglycan link proteins 1 and 4 have been shown to form dense, basket-like structures that surround the base of the IHCs (Kwiatkowska et al., 2016 [↗](#); Sonntag et al., 2018 [↗](#)). Furthermore, deletion of brevican results in alteration of pre- and postsynaptic spatial coupling at IHC synapses, indicating the possible role of extracellular matrix in transsynaptic organization. However, while the existing observations suggest that the synaptic changes in RTN4RL2 deficient animals might reflect derailed interaction of the SGN neurites with the extracellular matrix, the relevance of such an interaction of RTN4RL2 and Versican in the cochlea needs to be addressed in the future.

Materials and methods

Animals

RTN4RL2 null mutant mice on a C57BL/6N background were generated as previously described (Wörter et al., 2009 [↗](#)). Knock-out (RTN4RL2^{-/-}) and wild-type (RTN4RL2^{+/+}) mice were derived from heterozygous matings. Both male and female mice were used in this study. Animals were genotyped as previously described (Wörter et al., 2009 [↗](#)). The ages of the mice varied depending on the experiment as noted in the manuscript. The breeding and the experiments were approved by the Institutional Animal Care and Use Committee at the Medical University of Innsbruck, local Animal Welfare Committee of the University Medical Center Göttingen and the Max Planck Institute for Multidisciplinary Sciences, as well as the Animal Welfare Office of the state of Lower Saxony, Germany (LAVES, AZ: 19/3134). The ABR experiments were carried out under the approval of the Austrian Ministry of Education, Science and Research (reference number BMWFW-66.011/0120-WF/V/3b/2016).

Patch-clamp recordings

The apical turn of the organ of Corti was dissected from p21 to p29 animals in HEPES Hanks solution containing (in mM): 5.36 KCl, 141.7 NaCl, 10 HEPES, 0.5 MgSO₄, 1 MgCl₂, 5.6 D- glucose, and 3.4 L-glutamine (pH 7.2, ~ 300 mOsm/l). IHCs were exposed by gently removing of nearby supporting cells by negative pressure through a glass pipette from either modiolar or pillar side. All experiments were conducted at room temperature (RT, 20–25°C). Patch pipettes were made from GB150-8P or GB150F-8P borosilicate glass capillaries (Science Products, Hofheim, Germany) for perforated and ruptured patch-clamp configurations, respectively. Pipettes were fire polished

with a custom-made microforge and coated with Sylgard to minimize the capacitive noise, whenever capacitance recordings were performed. The measurements were performed using EPC-10 amplifier controlled by Patchmaster software (HEKA Elektronik, Germany). The holding potential for IHCs was set to -87 mV across all the experiments.

Ruptured patch-clamp

For ruptured patch-clamp we used extracellular solution containing (in mM): 2.8 KCl, 105 NaCl, 10 HEPES, 1 CsCl, 1 MgCl₂, 5 CaCl₂, 35 TEA-Cl, and 2 mg/ml D-glucose (pH 7.2, ~ 300 mOsm/l).

Intracellular solution contained in mM: 111 Cs-glutamate, 1 MgCl₂, 1 CaCl₂, 10 EGTA, 13 TEA-Cl, 20 HEPES, 4 Mg-ATP, 0.3 Na-GTP and 1 L-glutathione (pH 7.3, ~ 290 mOsm/l). Additionally, Ca²⁺ indicator Fluo4-FF (0.8 mM, Life Technologies) and TAMRA-conjugated RIBEYE/Ctbp2 binding peptide (10 mM, synthesized by the group of Dr. Olaf Jahn, Göttingen) were added to the intracellular solution for live imaging (Zenisek et al., 2004 [\[4\]](#)). Voltage dependency of Ca²⁺ influx was recorded by applying 20 ms long step depolarizations with 5 mV increment. For Ca²⁺ imaging, voltage ramp depolarizations ranging from -87 mV to 63 mV in the course of 150 ms were applied to the cells. Leak correction was performed using p/4 protocol and liquid junction potential of 17 mV was corrected offline. Recordings were discarded from the analysis if the series resistance (R_s) exceeded 14 MOhm during the first 3 minutes after rupturing the cell, leak current exceeded -50 pA at the holding potential and Ca²⁺ current rundown was more than 25%.

Recordings were analyzed using Igor Pro 6.3 (Wavemetrics) custom-written programs. Ca²⁺ current-voltage relationships (IV curves) were obtained by averaging approximately 5 ms segments at the maximal activation regions of individual current traces and plotting them against the depolarization voltages. Fractional activation curves of the Ca²⁺ channels were calculated from the IVs, normalized and fitted with the Boltzmann function:

$$G/G_{max} = \frac{1}{1 + \exp\left[\frac{V_m - V_{half}}{k}\right]}, \text{ where } G \text{ is conductance, } G_{max} \text{ is the maximal}$$

conductance, V_m is the membrane potential, V_{half} is the voltage corresponding to the half maximal activation of Ca²⁺ channels and k (slope of the curve) is the voltage sensitivity of Ca²⁺ channel activation.

Perforated patch clamp recordings

To perform simultaneous membrane capacitance (C_m) and Ca²⁺ current measurements we used extracellular solution containing (in mM): 106 NaCl, 35 TEA-Cl, 2.8 KCl, 1 MgCl₂, 1 CsCl, 10 HEPES, 3 CaCl₂, and 5.6 D-glucose (pH 7.2, ~ 300 mOsm/l). The pipette solution contained (in mM): 137 Cs-gluconate, 15 TEA-Cl, 3 GTP-Na₂, 1 ATP-Mg, 10 HEPES, 1 MgCl₂, as well as 300 mg/ml amphotericin B (pH 7.17, ~ 290 mOsm/l). Perforated patch-clamp recordings from IHCs were described previously (Moser and Beutner, 2000 [\[4\]](#)). C_m changes were measured using Lindau-Neher technique (Lindau and Neher, 1988 [\[4\]](#)). To evoke exocytosis IHCs were stimulated by step depolarizations (to -17 mV) of different durations (2, 5, 10, 20, 50, 100 ms) in randomized manner. Currents were leak-corrected using a p/5 protocol. Recordings were used only if the leak current was lower than 30 pA and the R_s was lower than 30 MOhm. Recordings were analyzed using Igor Pro 6.3 (Wavemetrics) custom-written programs. Ca²⁺ charge (Q_{Ca²⁺}) was calculated by the time integral of the leak-subtracted Ca²⁺ current during the depolarization step. ΔC_m was calculated as the difference between the average C_m before and after the depolarization. To measure average C_m we used 400 ms segments and skipped the initial 100 ms after the depolarization.

Ca²⁺ imaging

Ca²⁺ imaging was performed using spinning disc confocal microscope, as described before (Ohn et al., 2016 [\[4\]](#)). Briefly, the set-up was equipped with a spinning disc confocal unit (CSU22, Yokogawa) mounted on an upright microscope (Axio Examiner, Zeiss) and scientific CMOS camera (Andor

Neo). Images were acquired using 63x, 1.0 NA objective (W Plan-Apochromat, Zeiss). The pixel size was measured to be 103 nm.

IHCs were loaded with Fluo4-FF Ca^{2+} dye and TAMRA-conjugated Ctbp2 binding dimeric peptide via the patch pipette. First, the cells were scanned from bottom to top by imaging TAMRA fluorescence with 561 nm laser (Jive, Cobolt AB) and exposing each plane for 0.5 seconds. The stack was acquired with 0.5 μm step size using Piezo positioner (Piezosystem). This allowed us to identify the planes containing synaptic ribbons. Next, we recorded Fluo4-FF fluorescence increase at individual synapses by imaging ribbon containing planes with 491 nm laser (Calypso, Cobolt AB) at 100 Hz while applying voltage ramp depolarizations to the cell. 2 voltage ramps were applied at each plane, one being 5 ms shifted relative to the other. Spinning disk rotation speed was set to 2000 rpm, in order to synchronize with Ca^{2+} imaging. Images were analyzed using Igor Pro 6.3 (Wavemetrics). Ca^{2+} hotspots were identified by subtracting the average signal of several baseline frames from the average signal of 5 frames during stimulation (ΔF image). Since the same Ca^{2+} hotspot appears across multiple planes, the plane exhibiting the strongest signal was chosen. The intensities of the 3x3 matrix surrounding the central pixel of the hotspot were averaged across all time points to obtain the intensity profiles of Ca^{2+} influx over time. Afterwards, the background signal (average of approximately 60x60 pixel intensities outside the cell) was subtracted from the intensity-time profiles and $\Delta F/F_0$ traces were calculated. To enhance voltage resolution of Ca^{2+} imaging, two $\Delta F/F_0$ traces corresponding to two voltage ramp depolarizations (one shifted by 5 ms over the other) were combined and plotted against the corresponding voltages (FV curves). These curves were subsequently fitted with a modified Boltzmann function. Afterwards, fractional activation curves were calculated by fitting the linear decay of the fluorescence signal from the FV curves with a linear function (G_{max}) and then dividing the FV fit by the G_{max} line. The resulting curves were further fitted with a Boltzmann function. Maximal Ca^{2+} influx ($\Delta F/F_{0 \text{ max}}$) was calculated by averaging 5 points during the stimulation.

Immunohistochemistry

For whole mount immunofluorescence, cochleae were fixed in 4% formaldehyde on ice for 45-60 minutes or sometimes overnight. For anti- $\text{Ca}_v1.3$ stainings the cochleae were fixed for 10 minutes. After the fixation, the organs of Corti were microdissected in PBS and blocked in GSDB (goat serum dilution buffer; 16% goat serum, 20 mM phosphate buffer (PB), 0.3% Triton X-100, 0.45 M NaCl) for 1 hour at RT. The samples were then incubated in the primary antibody mixture overnight at 4°C. For anti-GluA4 immunolabelings samples were incubated at primary antibody mixture at 37°C overnight. The next day, samples were washed 3 times using wash buffer (20 mM PB, 0.3% Triton X-100, 0.45 M NaCl) followed by incubation in the secondary antibody mixture for 1 hour at RT. Afterwards, the samples were washed 3 times in the washing buffer, one time in PB and mounted using mounting medium (Mowiol).

The following primary antibodies were used (**Table 1**): rabbit anti-Homer1 (1:500, 160 002, Synaptic Systems), mouse anti-Ctbp2 (1:200, 612044, BD Biosciences), rabbit anti- $\text{Ca}_v1.3$ (1:100, ACC-005, Alomone Labs), mouse anti-Bassoon (1:300, ab82958, Abcam), mouse anti GluA2 (1:200, MAB397, Millipore), rabbit anti GluA4 (1:200, AB1508, Millipore), rabbit anti GluA2/3 (1:200, AB1506, Chemicon), rabbit anti-Myosin7a (1:800, ab3481, Abcam, or Proteus BioSciences), guinea pig anti-RibeyeA (1:500, 192104, Synaptic Systems), guinea pig anti- Synapsin1/2 (1:500, 106004, Synaptic Systems), guinea pig anti-Vglut3 (1:500, 135204, Synaptic Systems), chicken anti-parvalbumin (1:200, 195006, Synaptic Systems), chicken anti- calretinin (1:200, 214106, Synaptic Systems), guinea pig anti-VACHT (1:1000, 139105, Synaptic Systems), mouse anti-ATP1A3 (1:300, MA3-915, Thermo Fisher Scientific).

The following secondary antibodies were used: Alexa Fluor 488 conjugated anti-guinea pig (1:200, A11073, Thermo Fisher Scientific), Alexa Fluor 488 conjugated anti-rabbit (1:200, A11008, Thermo Fisher Scientific), Alexa Fluor 568 conjugated anti-chicken (1:200, ab175711, Abcam), Alexa Fluor

Antibody	Host species	Company	Dilution	Cat. No.	Validation
anti-Homer1	rabbit	Synaptic Systems	1:500	160002	Refer to manufacturer's datasheet
anti-Ctbp2	mouse	BD Biosciences	1:200	612044	KO validation; DOI: 10.7554/eLife.30241 DOI: 10.7554/eLife.29275
anti-Cav1.3	rabbit	Alomone Labs	1:100	ACC-005	KO validation; https://doi.org/10.1523/JNEUROSCI.3411-05.2005
anti-Bassoon	mouse	Abcam	1:300	ab8298	Refer to manufacturer's datasheet
anti GluA2	mouse	Millipore	1:200	MAB397	validated for use in ELISA, IC, IH, IP, RIA & WB; Refer to manufacturer's datasheet
anti-GluA2/3	rabbit	Chemicon	1:200	AB1506	validated for use in IH(P), IC, IH, IP, WB; Refer to manufacturer's datasheet
anti GluA4	rabbit	Millipore	1:200	AB1508	KO validation ; DOI: 10.1126/sciadv.aax5936
anti-Myosin7a	rabbit	Abcam/ Proteus BioSciences	1:800	ab3481	Refer to manufacturer's datasheet

Table 1.

List of primary antibodies.

anti-RibeyeA	guinea pig	Synaptic Systems	1:500	192104	Refer to manufacturer's datasheet
anti-Synapsin 1/2	guinea pig	Synaptic Systems	1:500,	106004	KO validation; Refer to manufacturer's datasheet
anti-Vglut3	guinea pig	Synaptic Systems	1:500	135204	KO validation; DOI: 10.3389/fncel.2017.00140
anti-Parvalbumin	chicken	Synaptic Systems	1:200	195006	Refer to manufacturer's datasheet
anti-Calretinin	chicken	Synaptic Systems	1:200	214106	Refer to manufacturer's datasheet
anti-VaChT	guinea pig	Synaptic Systems	1:1000	139105	Refer to manufacturer's datasheet
anti-ATP1A3	mouse	Invitrogen	1:300	MA3-915	Refer to manufacturer's datasheet
β III-tubulin	mouse	Biolegend	1:1000	801202	Refer to manufacturer's datasheet

Table 1. (continued)

633 conjugated anti-guinea pig (1:200, A21105, Thermo Fisher Scientific), STAR 580 conjugated anti-mouse (1:200, Abberior, ST635P-1001-500UG), STAR 635 conjugated anti-rabbit (1:200, Abberior, ST635P-1002-500UG).

Confocal stacks were acquired using Leica SP8 confocal microscope or Abberior Instruments Expert Line STED microscope.

The volumes of the Ctbp2 and Homer1 positive puncta were estimated using the surface algorithm of Imaris software (version 9.6.0, Bitplane). Following parameters were used to create the surfaces for both Ctbp2 and Homer1 puncta in all the analyzed stacks: surface detail 0.07 μm , background subtraction 0.562 μm , touching object size 0.4 μm .

The brightness and the contrast of the representative images were adjusted using Fiji (ImageJ) software for the visualization purposes.

Spiral ganglion neuron densities

RTN4RL2^{+/+} and RTN4RL2^{-/-} cochlea from p40 mice of either sex were processed for SGN counts. Mouse monoclonal anti- β III-tubulin (1:1000, 801202, Biolegend) at 4°C, followed by secondary Alexa Fluor 488 conjugated anti-mouse (1:1000, Thermo Fisher Scientific), was used to label all SGNs. Nuclei were visualized using a DAPI fluorescent counterstain (1:1000, Life Technologies). 20X images of mid-modiolar cochlear sections were collected. ImageJ was used to outline the spiral ganglion and generate the area. Quantitative assessment was performed on every other mid-modiolar section to reduce chances of double counting. Labeled cells were counted only if they had a round cell body, presence of nucleus, and homogenous cytoplasm. Densities were calculated and statistical differences were measured using a Student's t-test (GraphPad Prism, GraphPad Software Inc.).

Inner and outer Hair cell counts

IHC and OHC counts were performed for RTN4RL2^{+/+} and RTN4RL2^{-/-} mice at the ages of p15, 1 month and 2 months. Images were imported to ImageJ software, where IHC/OHC counts were performed blinded to genotype using the 'Cell counter' tool. To quantify the cells the nuclear staining of the DAPI channel was used. The number of IHCs and OHCs was assessed along a length of 100 μm for each image, and then averaged across all samples to obtain the average number of IHCs (or OHCs) per 100 μm .

RNAscope

RNA *in-situ* hybridization was performed using RNAscope[®] (Advanced Cell Diagnostics). Standard mouse probes were used to examine the expression of RTN4RL2 (450761, ACD) and Gria2 (865091, ACD) closely following the manufacturer's instructions. Briefly, 6 mm paraffin tissue sections underwent deparaffinization with xylene and a series of ethanol washes. Tissues were heated in kit-provided antigen retrieval buffer and digested by kit-provided proteinase. Sections were exposed to mFISH target probes and incubated at 40°C in a hybridization oven for 2 hours. After rinsing, mFISH signal was amplified using company-provided pre-amplifier and amplifier conjugated to fluorescent dye. Subsequently, sections were blocked with 1% BSA, 2% normal goat serum in 1xPBS containing 0.3% Triton X-100 for 1 hour at RT. The tissue was incubated in mouse anti- β III-tubulin antibody (1:500, 801202, Biolegend) overnight at 4°C. The next day, sections were rinsed with PBS, blocked again before incubating in secondary Alexa Fluor 488 conjugated anti-mouse antibody (1:1000, Invitrogen) for 2 hours at RT. Sections were counterstained with DAPI (1:1000, Life Technologies), mounted, and stored at 4°C until image analysis. mFISH images were captured on a Leica SP8 confocal microscope and processed using ImageJ.

Serial block-face scanning electron microscopy (SBEM)

Sample preparation for SBEM

One RTN4RL2^{+/+} and two RTN4RL2^{-/-} female mice at p36 were used for the SBEM experiments. Cochleae were processed for SBEM imaging as previously described (Hua et al., 2021 [DOI](#)).

In brief, the animals were decapitated after CO₂ inhalation under anesthesia. The cochleae were dissected from the skulls and immediately fixed by perfusing with ice-cold fixative through the round and oval windows at constant flow speed using an infusion pump (Micro4, WPI). The fixative solution was freshly prepared and made of 4% paraformaldehyde (Sigma- Aldrich), 2.5% glutaraldehyde (Sigma-Aldrich), and 0.08 M cacodylate (pH 7.4, Sigma-Aldrich). After being immersed in the fixative at 4°C for 5 hours, the cochleae were transferred to a decalcifying solution containing the same fixative and 5% ethylenediaminetetraacetic acid (EDTA, Serva) and incubated at 4°C for 5 hours. The samples were then washed twice with 0.15 M cacodylate (pH 7.4) for 30 min each, sequentially immersed in 2% OsO₄ (Sigma- Aldrich), 2.5% ferrocyanide (Sigma-Aldrich), and again 2% OsO₄ at RT for 2, 2, and 1.5 hours. After being washed in 0.15 M cacodylate and distilled water (Sartorius) for 30 min each, the samples were sequentially incubated in filtered 1% thiocarbohydrazide (TCH, Sigma-Aldrich) solution and 2% OsO₄ at RT for 1 and 2 hours, as well as in lead aspartate solution (0.03 M, pH 5.0, adjusted with KOH) at 50 °C for 2 hours with immediate two washing steps with distilled water at RT for 30 min each. For embedding, the samples were dehydrated through graded pre-cooled acetone (Carl Roth) series (50%, 75%, 90%, for 30 min each, all cooled at 4 °C) and then pure acetone at RT (three times for 30 min each). The sample infiltration started with 1:1 and 1:2 mixtures of acetone and Spurr's resin monomer (4.1 g ERL 4221, 0.95 g DER 736, 5.9 g NSA and 1% DMAE; Sigma-Aldrich) at RT for 6 and 12 hours on a rotator. After being impregnated in pure resin for 12 hours, the samples were placed in embedding molds (Polyscience) and hardened in a pre-warmed oven at 70°C for 72 hours.

Sample trimming and SBEM imaging

The sample blocks were mounted upright along the conical center axis on aluminum metal rivets (3VMRS12, Gatan, UK) and trimmed coronally towards the modiolus using a diamond trimmer (TRIM2, Leica, Germany). For each sample, a block face of about ~ 600 x 800 mm², centered at the apical segment based on the anatomical landmarks, was created using an ultramicrotome (UC7, Leica, Germany). The samples were coated with a 30 nm thick gold layer using a sputter coater (ACE600, Leica, Germany). The serial images were acquired using a field- emission scanning EM (Gemini300, Carl Zeiss, Germany) equipped with an in-chamber ultramicrotome (3ViewXP, Gatan, UK) and back-scattered electron detector (Onpoint, Gatan, UK). Focal charge compensation was set to 100 % with a high vacuum chamber pressure of 2.8 x 10³ mbar. The following parameters were set for the SBEM imaging: 12 nm pixel size, 50 nm nominal cutting thickness, 2 keV incident beam energy, and 1.5 ms pixel dwell time.

For the RTN4RL2^{+/+} dataset, 2377 consecutive slices (9000 × 15000 pixels) were collected, whereas the two RTN4RL2^{-/-} datasets had 3217 slices (16000 × 10000 pixels) and 2425 slices (9000 × 15000 pixels). All datasets were aligned along the z-direction using a custom MATLAB script based on cross-correlation maximum between consecutive slices (Hua et al., 2022 [DOI](#)) before being uploaded to webKnossos (Boergens et al., 2017 [DOI](#)) for skeleton and volume tracing.

Identification and quantification of auditory afferent fibers

In our SBEM datasets of the cochlea, manual skeleton tracing was carried out on all neurites that originated from three neighboring habenula perforata (HP) at the center of each dataset. To search for type I afferent fibers, several morphological features were used, such as myelination after

entering HP, radial and unbranched fiber trajectory, contact with IHCs, as well as ribbon-associated terminals (Hua et al., 2021 [DOI](#)). This resulted in 115 putative type I afferent fibers and further classification was made based on the presence of presynaptic ribbon, fiber branching, and contact with IHC. For an illustration purpose, a type I afferent fiber bundle of one RTN4RL2^{-/-} dataset was volume traced and 3D rendered using Amira software (Thermo Scientific, US).

Ribbon size measurement and synapse counting

Ribbon-type synapses were manually annotated in 18 intact IHCs captured by SBEM using webKnossos. The dense core region of individual ribbon synapses was manually contoured, and the associated voxels were counted for ribbon volume measurement. In the case of multi-ribbon synapses, all ribbon bodies at a single active zone were summed up to yield the ribbon volume.

Auditory brainstem responses

Auditory brainstem responses (ABR) were recorded in 2- to 4-month-old mice of either sex, as described previously (Luque et al., 2021 [DOI](#)). Briefly, we used a custom-made system in an anechoic chamber in a calibrated open-field configuration. ABRs were recorded via needle electrodes in response to tone bursts of 4, 8, 16, and 32 kHz or a click wide spectrum (2-45,2 KHz, 2 KHz steps) as stimuli. Tone pips of 3 ms duration (1 ms rise and fall time) were presented at a rate of 60/s with alternating phases. Starting with 0 dB, the stimuli were increased in 5 dB steps up to 120 dB. Hearing thresholds were determined as the minimum stimulation level that produced a clearly recognizable potential. Both ears were tested. Since there was no significant difference between the right and left ears, we did not consider this factor further. Recordings were evaluated by three independent researchers in a blinded manner.

Data analysis and statistics

The data were analyzed using Igor Pro (Wavemetrics), Python and GraphPad Prism (GraphPad Software Inc.) software. For 2 sample comparisons, data were tested for normality and equality of variances using Jarque-Bera and F-test, respectively. Afterwards, two-tailed Student's t-test or Mann-Whitney-Wilcoxon test were performed. The latter was used when normality and/or equality of variances were not met. P-values were corrected for multiple comparisons using Holm-Šidák method. For ABR thresholds Kruskal-Wallis test followed by Dunn's multiple comparison test was used. Data is presented as mean ± standard error of the mean (SEM), unless otherwise stated. Mean, SEM and standard deviation (SD) are indicated in the figure legends as mean ± SEM, SD. Significances are reported as *p < 0.05, **p < 0.01, ***p < 0.001. The number of the animals is indicated as N.

Acknowledgements

We thank Sina Langer, Christiane Senger-Freitag, Sandra Gerke for expert technical support. We thank Prof. Dr. Carolin Wichmann, Dr. Susann Michanski, Julius Bahr and Sophia Mutschall for the help with the SBEM sample preparation. We thank Dr. Yi Jiang and Haoyu Wang for the visualization of neurite reconstruction. We further thank Dr. Mark Rutherford for providing tips for GluA4 immunolabeling. We would also like to thank Prof. Dr. Olaf Jahn and Lars van Werven for the TAMRA-conjugated Ctb2 binding peptide synthesis. The study was supported by German Research Foundation through the Cluster of Excellence (EXC2067) Multiscale Bioimaging (TM) and the Leibniz Program (MO896/5 to TM) as well as by the European Research Council through the Advanced Grant “DynaHear” to TM under the European Union's Horizon 2020 Research and Innovation program (grant agreement No. 101054467), and by Fondation Pour l'Audition (FPA RD-2020-10), by the SPIN-FWF grant to CB, and by the National Natural Science Foundation of China (82171133 to YH), Industrial Support Fund of Huangpu District in Shanghai (XK2019011 to YH), Innovative Research Team of High-level Local Universities in Shanghai (SHSMU-ZLCX20211700).

NK is a member of the Hertha Sponer College from the Cluster of Excellence Multiscale Bioimaging (MBExC). TM is a Max-Planck Fellow at the Max Planck Institute for Multidisciplinary Sciences. Open access funding provided by Max Planck Society.

Additional information

Author contributions

CB, TM, YH, and NK designed the research. NK, MÜ, YQ, NB, FH, LJC, FW, ML, RG performed experiments and analyzed data. NK and TM wrote the paper with contributions of YH, CB, YQ, NB. CB, TM, YH acquired funding. All authors have reviewed and approved the final version of the manuscript for submission.

References

1. Appler J. M., Goodrich L. V (2011) **Connecting the ear to the brain: Molecular mechanisms of auditory circuit assembly** *Prog. Neurobiol* **93**:488–508 <https://doi.org/10.1016/j.pneurobio.2011.01.004>Google Scholar
2. Bäumer B. E., Kurz A., Borrie S. C., Sickinger S., Dours-Zimmermann M. T., Zimmermann D. R., et al. (2014) **Nogo receptor homolog NgR2 expressed in sensory DRG neurons controls epidermal innervation by interaction with Versican** *J. Neurosci. Off. J. Soc. Neurosci* **34**:1633–1646 <https://doi.org/10.1523/JNEUROSCI.3094-13.2014>Google Scholar
3. Boergens K. M., Berning M., Bocklisch T., Bräunlein D., Drawitsch F., Frohnhofen J., et al. (2017) **webKnossos: efficient online 3D data annotation for connectomics** *Nat. Methods* **14**:691–694 <https://doi.org/10.1038/nmeth.4331>Google Scholar
4. Borrie S. C., Sartori S. B., Lehmann J., Sah A., Singewald N., Bandtlow C. E. (2014) **Loss of Nogo receptor homolog NgR2 alters spine morphology of CA1 neurons and emotionality in adult mice** *Front. Behav. Neurosci.* <https://www.frontiersin.org/articles/10.3389/fnbeh.2014.00175>
5. Bulankina A. V., Moser T (2012) **Neural circuit development in the mammalian cochlea** *Physiology* **27**:100–112 <https://doi.org/10.1152/physiol.00036.2011>Google Scholar
6. Cantu-Guerra H. L., Papazian M. R., Gorsky A. L., Alekos N. S., Caccavano A., Karagulyan N., et al. (2023) **Cochlear hair cell innervation is dependent on a modulatory function of SEMAPHORIN-3A** *Dev. Dyn* **252**:124–144 <https://doi.org/10.1002/dvdy.548>Google Scholar
7. Carlton A. J., Jeng J.-Y., Grandi F. C., De Faveri F., Amariutei A. E., De Tomasi L., et al. (2024) **BAI1 localizes AMPA receptors at the cochlear afferent post-synaptic density and is essential for hearing** *Cell Rep* **43**:114025 <https://doi.org/10.1016/j.celrep.2024.114025>Google Scholar
8. Coate T. M., Scott M. K., Gurjar M (2019) **Current concepts in cochlear ribbon synapse formation** *Synapse* **73**:e22087 <https://doi.org/10.1002/syn.22087>Google Scholar
9. Coate T. M., Spita N. A., Zhang K. D., Isgrig K. T., Kelley M. W (2015) **Neuropilin- 2/Semaphorin-3F-mediated repulsion promotes inner hair cell innervation by spiral ganglion neurons** *eLife* **4** <https://doi.org/10.7554/eLife.07830>Google Scholar
10. Defourny J., Poirrier A.-L., Lallemand F., Mateo Sánchez S., Neef J., Vanderhaeghen P., et al. (2013) **Ephrin-A5/EphA4 signalling controls specific afferent targeting to cochlear hair cells** *Nat. Commun* **4**:1438 <https://doi.org/10.1038/ncomms2445>Google Scholar
11. Domeniconi M., Cao Z., Spencer T., Sivasankaran R., Wang K. C., Nikulina E., et al. (2002) **Myelin-Associated Glycoprotein Interacts with the Nogo66 Receptor to Inhibit Neurite Outgrowth** *Neuron* **35**:283–290 [https://doi.org/10.1016/S0896-6273\(02\)00770-5](https://doi.org/10.1016/S0896-6273(02)00770-5)Google Scholar

12. Elkon R., Milon B., Morrison L., Shah M., Vijayakumar S., Racherla M., et al. (2015) **RFX transcription factors are essential for hearing in mice** *Nat. Commun* **6**:8549 <https://doi.org/10.1038/ncomms9549>Google Scholar
13. Fell B., Eckrich Stephanie, Hecker Dietmar, Obermair Gerald J., Flockerzi V., Schick B., et al. (2015) **$\alpha 2\delta 2$ Controls the Function and Transsynaptic Coupling of Cav1.3 Channels in Mouse Inner Hair Cells and is Essential for Normal Hearing** *J Neurosci.* **36**:11024–11036 [Google Scholar](#)
14. Fournier A. E., GrandPre T., Strittmatter S. M (2001) **Identification of a receptor mediating Nogo-66 inhibition of axonal regeneration** *Nature* **409**:341–346 <https://doi.org/10.1038/35053072>Google Scholar
15. Hua Y., Ding X., Wang H., Wang F., Lu Y., Neef J., et al. (2021) **Electron Microscopic Reconstruction of Neural Circuitry in the Cochlea** *Cell Rep* **34**:108551 <https://doi.org/10.1016/j.celrep.2020.108551>Google Scholar
16. Hua Y., Loomba S., Pawlak V., Voit K.-M., Laserstein P., Boergens K. M., et al. (2022) **Connectomic analysis of thalamus-driven disinhibition in cortical layer 4** *Cell Rep* **41**:111476 <https://doi.org/10.1016/j.celrep.2022.111476>Google Scholar
17. Jaime Tobón L. M., Moser T. (2023a) **Bridging the gap between presynaptic hair cell function and neural sound encoding** *eLife* **12**:RP93749 <https://doi.org/10.7554/eLife.93749>Google Scholar
18. Jaime Tobón L. M., Moser T. (2023b) **Ca²⁺ regulation of glutamate release from inner hair cells of hearing mice** *Proc. Natl. Acad. Sci* **120**:e2311539120 <https://doi.org/10.1073/pnas.2311539120>Google Scholar
19. Jean P., de la Morena D. L., Michanski S., Tobón L. M. J., Chakrabarti R., Picher M. M., et al. (2018) **The synaptic ribbon is critical for sound encoding at high rates and with temporal precision** *eLife* **7**:e29275 <https://doi.org/10.7554/eLife.29275>Google Scholar
20. Jean P., Tai Wong Jun, Singh-Estivalet F., Lelli A., Scandola A., Megharba C., et al. (2023) **Single-cell transcriptomic profiling of the mouse cochlea: An atlas for targeted therapies** *Proc. Natl. Acad. Sci* **120**:e2221744120 <https://doi.org/10.1073/pnas.2221744120>Google Scholar
21. Johnson S. L., Safieddine S., Mustapha M., Marcotti W (2019) **Hair Cell Afferent Synapses: Function and Dysfunction** *Cold Spring Harb. Perspect. Med* **9** <https://doi.org/10.1101/cshperspect.a033175>Google Scholar
22. Jukic A., Lei Z., Cebul E. R., Pinter K., Tadesse Y., Jarysta A., et al. (2024) **Presynaptic Nrnx3 is essential for ribbon-synapse maturation in hair cells** *Development* **151**:dev202723 <https://doi.org/10.1242/dev.202723>Google Scholar
23. Jung J. S., Zhang K. D., Wang Z., McMurray M., Tkaczuk A., Ogawa Y., et al. (2019) **Semaphorin-5B Controls Spiral Ganglion Neuron Branch Refinement during Development** *J. Neurosci* **39**:6425–6438 <https://doi.org/10.1523/JNEUROSCI.0113-19.2019>Google Scholar
24. Karagulyan N., Thirumalai A., Michanski S., Qi Y., Fang Q., Wang H., et al. (2025) **Gating of hair cell Ca²⁺ channels governs the activity of cochlear neurons** *bioRxiv* :2025.01.23.634330 <https://doi.org/10.1101/2025.01.23.634330>Google Scholar

25. Koundakjian E. J., Appler J. L., Goodrich L. V (2007) **Auditory neurons make stereotyped wiring decisions before maturation of their targets** *J. Neurosci. Off. J. Soc. Neurosci* **27**:14078–14088 <https://doi.org/10.1523/JNEUROSCI.3765-07.2007>Google Scholar
26. Kujawa S. G., Liberman M. C (2009) **Adding insult to injury: cochlear nerve degeneration after “temporary” noise-induced hearing loss** *J. Neurosci* **29**:14077 [Google Scholar](#)
27. Kujawa S. G., Liberman M. C (2015) **Synaptopathy in the noise-exposed and aging cochlea: Primary neural degeneration in acquired sensorineural hearing loss** *Hear. Res* **330**:191–199 <https://doi.org/10.1016/j.heares.2015.02.009>Google Scholar
28. Kwiatkowska M., Reinhard J., Roll L., Kraft N., Dazert S., Faissner A., et al. (2016) **The expression pattern and inhibitory influence of Tenascin-C on the growth of spiral ganglion neurons suggest a regulatory role as boundary formation molecule in the postnatal mouse inner ear** *Neuroscience* **319**:46–58 <https://doi.org/10.1016/j.neuroscience.2016.01.039>Google Scholar
29. Liberman L. D., Liberman M. C (2016) **Postnatal maturation of auditory-nerve heterogeneity, as seen in spatial gradients of synapse morphology in the inner hair cell area** *Hear. Res* **339**:12–22 <https://doi.org/10.1016/j.heares.2016.06.002>Google Scholar
30. Lindau M., Neher E (1988) **Patch-clamp techniques for time-resolved capacitance measurements in single cells** *Pflüg. Arch. Eur. J. Physiol* **411**:137–146 [Google Scholar](#)
31. Liu H., Chen L., Giffen K. P., Stringham S. T., Li Y., Judge P. D., et al. (2018) **Cell-Specific Transcriptome Analysis Shows That Adult Pillar and Deiters’ Cells Express Genes Encoding Machinery for Specializations of Cochlear Hair Cells** *Front. Mol. Neurosci* **11** <https://doi.org/10.3389/fnmol.2018.00356>Google Scholar
32. Lu Y., Liu J., Li B., Wang H., Wang F., Wang S., et al. (2024) **Spatial patterns of noise- induced inner hair cell ribbon loss in the mouse mid-cochlea** *iScience* **27**:108825 <https://doi.org/10.1016/j.isci.2024.108825>Google Scholar
33. Luque M., Schrott-Fischer A., Dudas J., Pechriggl E., Brenner E., Rask-Andersen H., et al. (2021) **HCN channels in the mammalian cochlea: Expression pattern, subcellular location, and age-dependent changes** *J. Neurosci. Res* **99**:699–728 <https://doi.org/10.1002/jnr.24754>Google Scholar
34. Matsubara A., Laake J. H., Davanger S., Usami S., Ottersen O. P (1996) **Organization of AMPA receptor subunits at a glutamate synapse: a quantitative immunogold analysis of hair cell synapses in the rat organ of Corti** *J. Neurosci* **16**:4457 [Google Scholar](#)
35. Meyer A. C., Moser T (2010) **Structure and function of cochlear afferent innervation** *Curr. Opin. Otolaryngol. Head Neck Surg* **18**:441–446 <https://doi.org/10.1097/MOO.0b013e32833e0586>Google Scholar
36. Michanski S., Smaluch K., Steyer A. M., Chakrabarti R., Setz C., Oestreicher D., et al. (2019) **Mapping developmental maturation of inner hair cell ribbon synapses in the apical mouse cochlea** *Proc. Natl. Acad. Sci* **116**:6415–6424 <https://doi.org/10.1073/pnas.1812029116>Google Scholar

37. Mironova Y. A., Giger R. J (2013) **Where no synapses go: gatekeepers of circuit remodeling and synaptic strength** *Trends Neurosci* **36**:363–373 <https://doi.org/10.1016/j.tins.2013.04.003>Google Scholar
38. Moser T., Beutner D (2000) **Kinetics of exocytosis and endocytosis at the cochlear inner hair cell afferent synapse of the mouse** *Proc. Natl. Acad. Sci* **97**:883–888 <https://doi.org/10.1073/pnas.97.2.883>Google Scholar
39. Moverman D. J., Liberman L. D., Kraemer S., Corfas G., Liberman M. C (2023) **Ultrastructure of noise-induced cochlear synaptopathy** *Sci. Rep* **13**:19456 <https://doi.org/10.1038/s41598-023-46859-6>Google Scholar
40. Niedzielski A. S., Wenthold R. J (1995) **Expression of AMPA, kainate, and NMDA receptor subunits in cochlear and vestibular ganglia** *J. Neurosci* **15**:2338 [Google Scholar](https://doi.org/10.1523/JNEUROSCI.15-08.1995)
41. Ohn T.-L., Rutherford M. A., Jing Z., Jung S., Duque-Afonso C. J., Hoch G., et al. (2016) **Hair cells use active zones with different voltage dependence of Ca²⁺ influx to decompose sounds into complementary neural codes** *Proc. Natl. Acad. Sci* **113**:E4716–E4725 <https://doi.org/10.1073/pnas.1605737113>Google Scholar
42. Özçete Ö. D., Moser T (2021) **A sensory cell diversifies its output by varying Ca²⁺ influx-release coupling among active zones** *EMBO J* **40**:e106010 <https://doi.org/10.15252/embj.2020106010>Google Scholar
43. Ramirez M. A., Ninoyu Y., Miller C., Andrade L. R., Edassery S., Bomba-Warczak E., et al. (2022) **Cochlear ribbon synapse maturation requires Nlgn1 and Nlgn3** *iScience* **25**:104803 <https://doi.org/10.1016/j.isci.2022.104803>Google Scholar
44. Roux I., Safieddine S., Nouvian R., Grati M., Simmler M.-C., Bahloul A., et al. (2006) **Otoferlin, Defective in a Human Deafness Form, Is Essential for Exocytosis at the Auditory Ribbon Synapse** *Cell* **127**:277–289 <https://doi.org/10.1016/j.cell.2006.08.040>Google Scholar
45. Ruel J., Emery S., Nouvian R., Bersot T., Amilhon B., Van Rybroek J. M., et al. (2008) **Impairment of SLC17A8 Encoding Vesicular Glutamate Transporter-3, VGLUT3, Underlies Nonsyndromic Deafness DFNA25 and Inner Hair Cell Dysfunction in Null Mice** *Am. J. Hum. Genet.* **83**:278–292 <https://doi.org/10.1016/j.ajhg.2008.07.008>Google Scholar
46. Rutherford M. A., Bhattacharyya A., Xiao M., Cai H.-M., Pal I., Rubio M. E (2023) **GluA3 subunits are required for appropriate assembly of AMPAR GluA2 and GluA4 subunits on cochlear afferent synapses and for presynaptic ribbon modiolar-pillar morphology** *eLife* **12**:e80950 <https://doi.org/10.7554/eLife.80950>Google Scholar
47. Schwab M. E (2010) **Functions of Nogo proteins and their receptors in the nervous system** *Nat. Rev. Neurosci* **11**:799–811 <https://doi.org/10.1038/nrn2936>Google Scholar
48. Seal R. P., Akil O., Yi E., Weber C. M., Grant L., Yoo J., et al. (2008) **Sensorineural Deafness and Seizures in Mice Lacking Vesicular Glutamate Transporter 3** *Neuron* **57**:263–275 <https://doi.org/10.1016/j.neuron.2007.11.032>Google Scholar
49. Sherrill H. E., Jean P., Driver E. C., Sanders T. R., Fitzgerald T. S., Moser T., et al. (2019) **Pou4f1 Defines a Subgroup of Type I Spiral Ganglion Neurons and Is Necessary for Normal Inner**

Hair Cell Presynaptic Ca²⁺ Signaling *J. Neurosci* **39**:5284–5298 <https://doi.org/10.1523/JNEUROSCI.2728-18.2019>Google Scholar

50. Shrestha B. R., Chia C., Wu L., Kujawa S. G., Liberman M. C., Goodrich L. V (2018) **Sensory Neuron Diversity in the Inner Ear Is Shaped by Activity** *Cell* **174**:1229–1246 <https://doi.org/10.1016/j.cell.2018.07.007>Google Scholar
51. Shrestha B. R., Wu L., Goodrich L. V (2023) **Runx1 controls auditory sensory neuron diversity in mice** *Dev. Cell* **58**:306–319 <https://doi.org/10.1016/j.devcel.2023.01.008>Google Scholar
52. Sonntag M., Blosa M., Schmidt S., Reimann K., Blum K., Eckrich T., et al. (2018) **Synaptic coupling of inner ear sensory cells is controlled by brevicin-based extracellular matrix baskets resembling perineuronal nets** *BMC Biol* **16**:99 <https://doi.org/10.1186/s12915-018-0566-8>Google Scholar
53. Stephenson J. R., Paavola K. J., Schaefer S. A., Kaur B., Van Meir E. G., Hall R. A. (2013) **Brain-specific Angiogenesis Inhibitor-1 Signaling, Regulation, and Enrichment in the Postsynaptic Density** *J. Biol. Chem* **288**:22248–22256 <https://doi.org/10.1074/jbc.M113.489757>Google Scholar
54. Vaccaro G., Dumoulin A., Zuñiga N. R., Bandtlow C. E., Stoeckli E. T (2022) **The Nogo- 66 Receptors NgR1 and NgR3 Are Required for Commissural Axon Pathfinding** *J. Neurosci* **42**:4087–4100 <https://doi.org/10.1523/JNEUROSCI.1390-21.2022>Google Scholar
55. Venkatesh K., Chivatakarn O., Lee H., Joshi P. S., Kantor D. B., Newman B. A., et al. (2005) **The Nogo-66 Receptor Homolog NgR2 Is a Sialic Acid-Dependent Receptor Selective for Myelin-Associated Glycoprotein** *J. Neurosci* **25**:808–822 <https://doi.org/10.1523/JNEUROSCI.4464-04.2005>Google Scholar
56. Wang J., Miao Y., Wicklein R., Sun Z., Wang J., Jude K. M., et al. (2021) **RTN4/NoGo- receptor binding to BAI adhesion-GPCRs regulates neuronal development** *Cell* **184**:5869–5885 <https://doi.org/10.1016/j.cell.2021.10.016>Google Scholar
57. Wang K. C., Koprivica V., Kim J. A., Sivasankaran R., Guo Y., Neve R. L., et al. (2002a) **Oligodendrocyte-myelin glycoprotein is a Nogo receptor ligand that inhibits neurite outgrowth** *Nature* **417**:941–944 <https://doi.org/10.1038/nature00867>Google Scholar
58. Wang X., Chun S.-J., Treloar H., Vartanian T., Greer C. A., Strittmatter S. M (2002b) **Localization of Nogo-A and Nogo-66 Receptor Proteins at Sites of Axon-Myelin and Synaptic Contact** *J. Neurosci* **22**:5505–5515 <https://doi.org/10.1523/JNEUROSCI.22-13-05505.2002>Google Scholar
59. Wills Z. P., Mandel-Brehm C., Mardinly A. R., McCord A. E., Giger R. J., Greenberg M. E (2012) **The Nogo Receptor Family Restricts Synapse Number in the Developing Hippocampus** *Neuron* **73**:466–481 <https://doi.org/10.1016/j.neuron.2011.11.029>Google Scholar
60. Wong A. B., Rutherford M. A., Gabrielaitis M., Pangršič T., Göttfert F., Frank T., et al. (2014) **Developmental refinement of hair cell synapses tightens the coupling of Ca²⁺ influx to exocytosis** *EMBO J* **33**:247–264 <https://doi.org/10.1002/embj.201387110>Google Scholar
61. Wörter V., Schweigreiter R., Kinzel B., Mueller M., Barske C., Böck G., et al. (2009) **Inhibitory activity of myelin-associated glycoprotein on sensory neurons is largely independent of**

NgR1 and NgR2 and resides within Ig-Like domains 4 and 5 *PloS One* **4**:e5218 <https://doi.org/10.1371/journal.pone.0005218> [Google Scholar](#)

62. Zenisek D., Horst N. K., Merrifield C., Sterling P., Matthews G (2004) **Visualizing synaptic ribbons in the living cell** *J. Neurosci. Off. J. Soc. Neurosci* **24**:9752–9759 <https://doi.org/10.1523/JNEUROSCI.2886-04.2004> [Google Scholar](#)

Author information

Nare Karagulyan[#]

Institute for Auditory Neuroscience, University Medical Center Göttingen, Göttingen, Germany, Auditory Neuroscience and Synaptic Nanophysiology Group, Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany, Cluster of Excellence “Multiscale Bioimaging of Excitable Cells”, Göttingen, Germany
ORCID iD: [0009-0005-3999-2427](https://orcid.org/0009-0005-3999-2427)

[#]equal contribution

Maja Überegger[#]

Institute of Neurobiochemistry, Biocenter, Medical University of Innsbruck, Innsbruck, Austria

[#]equal contribution

Yumeng Qi[#]

Shanghai Institute of Precision Medicine, Ninth People’s Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

[#]equal contribution

Norbert Babai

Institute for Auditory Neuroscience, University Medical Center Göttingen, Göttingen, Germany, Auditory Neuroscience and Synaptic Nanophysiology Group, Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany

Florian Hofer

Institute of Neurobiochemistry, Biocenter, Medical University of Innsbruck, Innsbruck, Austria

Lejo Johnson Chacko

Department of Otorhinolaryngology, Medical University of Innsbruck, Innsbruck, Austria

Fangfang Wang

Shanghai Institute of Precision Medicine, Ninth People’s Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

Maria Luque

Department of Otorhinolaryngology, Medical University of Innsbruck, Innsbruck, Austria

Rudolf Glueckert

Department of Otorhinolaryngology, Medical University of Innsbruck, Innsbruck, Austria

Anneliese Schrott-Fischer

Department of Otorhinolaryngology, Medical University of Innsbruck, Innsbruck, Austria

Yunfeng Hua

Shanghai Institute of Precision Medicine, Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

For correspondence: yunfeng.hua@shsmu.edu.cn

Tobias Moser

Institute for Auditory Neuroscience, University Medical Center Göttingen, Göttingen, Germany, Auditory Neuroscience and Synaptic Nanophysiology Group, Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany, Cluster of Excellence “Multiscale Bioimaging of Excitable Cells”, Göttingen, Germany, Department for Hearing, Speech and Voice Disorders, Medical University of Innsbruck, Innsbruck, Austria
ORCID iD: [0000-0001-7145-0533](https://orcid.org/0000-0001-7145-0533)

For correspondence: tmoser@gwdg.de

Christine Bandtlow

Institute of Neurobiochemistry, Biocenter, Medical University of Innsbruck, Innsbruck, Austria

For correspondence: christine.bandtlow@i-med.ac.at

Editors

Reviewing Editor

Annalisa Scimemi

University at Albany, State University of New York, Albany, United States of America

Senior Editor

Andrew King

University of Oxford, Oxford, United Kingdom

Reviewer #1 (Public review):

Hearing and balance rely on specialized ribbon synapses that transmit sensory stimuli between hair cells and afferent neurons. Synaptic adhesion molecules that form and regulate transsynaptic interactions between inner hair cells (IHCs) and spiral ganglion neurons (SGNs) are crucial for maintaining auditory synaptic integrity and, consequently, for auditory signaling. Synaptic adhesion molecules such as neurexin-3 and neuroligin-1 and -3 have recently been shown to play vital roles in establishing and maintaining these synaptic connections (doi: 10.1242/dev.202723 and DOI: 10.1016/j.isci.2022.104803). However, the full set of molecules required for synapse assembly remains unclear.

Karagulan et al. highlight the critical role of the synaptic adhesion molecule RTN4RL2 in the development and function of auditory afferent synapses between IHCs and SGNs, particularly regarding how RTN4RL2 may influence synaptic integrity and receptor localization. Their study shows that deletion of RTN4RL2 in mice leads to enlarged presynaptic ribbons and smaller postsynaptic densities (PSDs) in SGNs, indicating that RTN4RL2 is vital for synaptic structure. Additionally, the presence of "orphan" PSDs—those not directly associated with IHCs—in RTN4RL2 knockout mice suggests a developmental defect in which some SGN neurites fail to form appropriate synaptic contacts, highlighting potential issues in synaptic pruning or guidance. The study also observed a depolarized shift in the activation of CaV1.3 calcium channels in IHCs, indicating altered presynaptic functionality that may lead to impaired neurotransmitter release. Furthermore, postsynaptic SGNs exhibited a deficiency in GluA2/3 AMPA receptor subunits, despite normal Gria2 mRNA levels, pointing to a disruption in receptor localization that could compromise synaptic transmission. Auditory brainstem responses showed increased sound thresholds in RTN4RL2 knockout mice, indicating impaired hearing related to these synaptic dysfunctions.

The findings reported here significantly enhance our understanding of synaptic organization in the auditory system, particularly concerning the molecular mechanisms underlying IHC-SGN connectivity. The implications are far-reaching, as they not only inform auditory neuroscience but also provide insights into potential therapeutic targets for hearing loss related to synaptic dysfunction.

Comments on the Latest Version:

In the revised manuscript, the authors have addressed my previous comments and incorporated my recommendations by adding missing experimental details, using color-blind-friendly figure colors, and discussing the differences between GluA3 KO and RTN4RL2 KO phenotypes. They also clarified why the animals needed for additional experiments are no longer available. Although these specific animals are unavailable, the authors made an effort to address my concerns by performing

<https://doi.org/10.7554/eLife.103481.2.sa2>

Reviewer #3 (Public review):

In this study, the authors used RNAscope to explore the expression of RTN4RL2 RNA in hair cells and spiral ganglia. Through RTN4RL2 gene knockout mice, they demonstrated that the absence of RTN4RL2 leads to pre-synaptic changes of an increase in the size of presynaptic ribbons and a depolarized shift in the activation of calcium channels in inner hair cells. Additionally, they observed a post-synaptic reduction in GluA2-4 AMPA receptors and identified additional "orphan PSDs" not paired with presynaptic ribbons via immunostaining and an increased number of type I SGNs that are not connected with a ribbon synapse via serial block face imaging. These synaptic alterations ultimately resulted in an increased hearing threshold in mice, confirming that the RTN4RL2 gene is essential for normal hearing. These data are intriguing as they suggest that RTN4RL2 contributes to the proper formation and function of auditory afferent synapses and is critical for normal hearing. Most strikingly, the post-synaptic changes and hearing threshold changes are similar to recently published results by Carlton et al, 2024 on a mutation in *Bai1*, which is a potential binding partner for RTN4RL2. Overall this work provides some clues to the function of RTN4RL2 in the cochlea, but further studies are required to elucidate the function.

A few points would improve the manuscript and the strength of the data presented.

(1) A quantitative assessment is necessary in Figure 1 when discussing RNA scope data. It would be beneficial to show that expression levels are quantitatively reduced in KO mice compared to wild-type mice. This suggestion also applies to Figure 3D, which examines expression levels of Gria2. Data is provided for KO reduction in SGN, but not showing that hair cell labeling is specific. If slides are not available for the young ages, showing hair cell expression at P40 would be sufficient along with a loss of labeling at in the KO at P40.

(2) In Figure 2, the authors present a morphological analysis of synapses and discuss the presence of "orphan PSDs." I agree that Homer1 not juxtaposed with Ctbp2 is increased in KO mice compared to the control group. However, in quantifying this, they opted to measure the number of Ctbp2 puncta with Homer 1 juxtaposed, which indicates the percentages of orphan ribbons rather than directly quantifying the number of Homer1 not juxtaposed with Ctbp2. Quantifying the number of Homer1 not juxtaposed with Ctbp2 would more clearly represent "orphan PSDs" and provide stronger support for the discussion surrounding their presence. A measurement of these was provided in the rebuttal letter, and while this number much more clearly demonstrates the increase in the number of orphan puncta, this analysis is not provided in the manuscript. This number also suggests the number of orphan receptors may be quite high, outnumbering ribbons 2:1.

(3) In Figure 3, the authors discuss GluA2/3 puncta reduction and note that Gria2 RNA expression remains unchanged. However, the GluA2/3 labeling is done at 1-1.5 months, whereas the Gria2 RNAscope is done at P4. Additionally, there is a lack of quantification for Gria2 RNA expression due to their tissue being processed separately. RNA scope at a comparable age to the GluA2/3 would be stronger support for their statement that Gria2 expression is comparable despite a reduction in GluA2/3 puncta.

(4) In Figure 4, the authors indicate that RTN4RL2 deficiency reduces the number of type 1 SGNs connected to ribbons. Given that the number of ribbons remains unchanged (Figure 2), it is important to clearly explain the implications of this finding. It is already known that each type I SGN forms a single synaptic contact with a single IHC. The fact that the number of ribbons remains constant while additional "orphan PSDs" are present suggests that the overall number of SGNs might need to increase to account for these findings, however, the authors noted no change in the number of SGN soma. This discrepancy is important to point out.

<https://doi.org/10.7554/eLife.103481.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Hearing and balance rely on specialized ribbon synapses that transmit sensory stimuli between hair cells and afferent neurons. Synaptic adhesion molecules that form and regulate transsynaptic interactions between inner hair cells (IHCs) and spiral ganglion neurons (SGNs) are crucial for maintaining auditory synaptic integrity and, consequently, for auditory signaling. Synaptic adhesion molecules such as neuroligin-3 and neuroligin-1 and -3 have recently been shown to play vital roles in establishing and maintaining these synaptic connections (doi: 10.1242/dev.202723 and DOI: 10.1016/j.isci.2022.104803). However, the full set of molecules required for synapse assembly remains unclear.

Karagulan et al. highlight the critical role of the synaptic adhesion molecule RTN4RL2 in the development and function of auditory afferent synapses between IHCs and SGNs, particularly regarding how RTN4RL2 may influence synaptic integrity and receptor localization. Their study shows that deletion of RTN4RL2 in mice leads to enlarged presynaptic ribbons and smaller postsynaptic densities (PSDs) in SGNs, indicating that RTN4RL2 is vital for synaptic structure. Additionally, the presence of "orphan" PSDs—those not directly associated with IHCs—in RTN4RL2 knockout mice suggests a developmental defect in which some SGN neurites fail to form appropriate synaptic contacts, highlighting potential issues in synaptic pruning or guidance. The study also observed a depolarized shift in the activation of CaV1.3 calcium channels in IHCs, indicating altered presynaptic functionality that may lead to impaired neurotransmitter release. Furthermore, postsynaptic SGNs exhibited a deficiency in GluA2/3 AMPA receptor subunits, despite normal Gria2 mRNA levels, pointing to a disruption in receptor localization that could compromise synaptic transmission. Auditory brainstem responses showed increased sound thresholds in RTN4RL2 knockout mice, indicating impaired hearing related to these synaptic dysfunctions.

The findings reported here significantly enhance our understanding of synaptic organization in the auditory system, particularly concerning the molecular mechanisms underlying IHC-SGN connectivity. The implications are far-reaching, as they not only inform auditory neuroscience but also provide insights into potential therapeutic targets for hearing loss related to synaptic dysfunction.

We would like to thank the reviewer for appreciating the work and the advice that helped us to further improve the manuscript. We have carefully addressed all concerns, please see our point-per-point response below and the revised manuscript.

Reviewer #2 (Public review):

Summary:

Kargulyan et al. investigate the function of the transsynaptic adhesion molecule RTN4RL2 in the formation and function of ribbon synapses between type I spiral ganglion neurons (SGNs) and inner hair cells. For this purpose, they study constitutive RTN4RL2 knock-out mice. Using immunohistochemistry, they reveal defects in the recruitment of protein to ribbon synapses in the knockouts. Serial block phase EM reveals defects in SGN projections in mutants. Electrophysiological recordings suggest a small but statistically significant depolarized shift in the activation of Cav1.3 Ca^{2+} channels. Auditory thresholds are also elevated in the mutant mice. The authors conclude that RTN4RL2 contributes to the formation and function of auditory afferent synapses to regulate auditory function.

We would like to thank the reviewer for appreciating the work and the advice that helped us to further improve the manuscript. We have carefully addressed all concerns, please see our point-per-point response below and the revised manuscript.

Strengths:

The authors have excellent tools to analyze ribbon synapses.

Weaknesses:

However, there are several concerns that substantially reduce my enthusiasm for the study.

(1) The analysis of the expression pattern of RTN4RL2 in Figure 1 is incomplete. The authors should show a developmental time course of expression up into maturity to correlate gene expression with major developmental milestones such as axon outgrowth, innervation, and refinement. This would allow the development of models supporting roles in axon outgrowth versus innervation or both.

We agree that it would be valuable to show the developmental time course of RTN4RL2 expression. In response to the reviewer's comment, we are providing RNAscope data from developmental ages E11.5, E12.5 and E16 in Figure 1. RTN4RL2 shows expression at E11.5/E12.5 both in the spiral ganglion and hair cell region, with first onset in the hair cells. We conclude that RTN4RL2 is expressed highest during fiber growth at embryonic stages and is downregulated during postnatal development maintaining low levels of expression during adulthood.

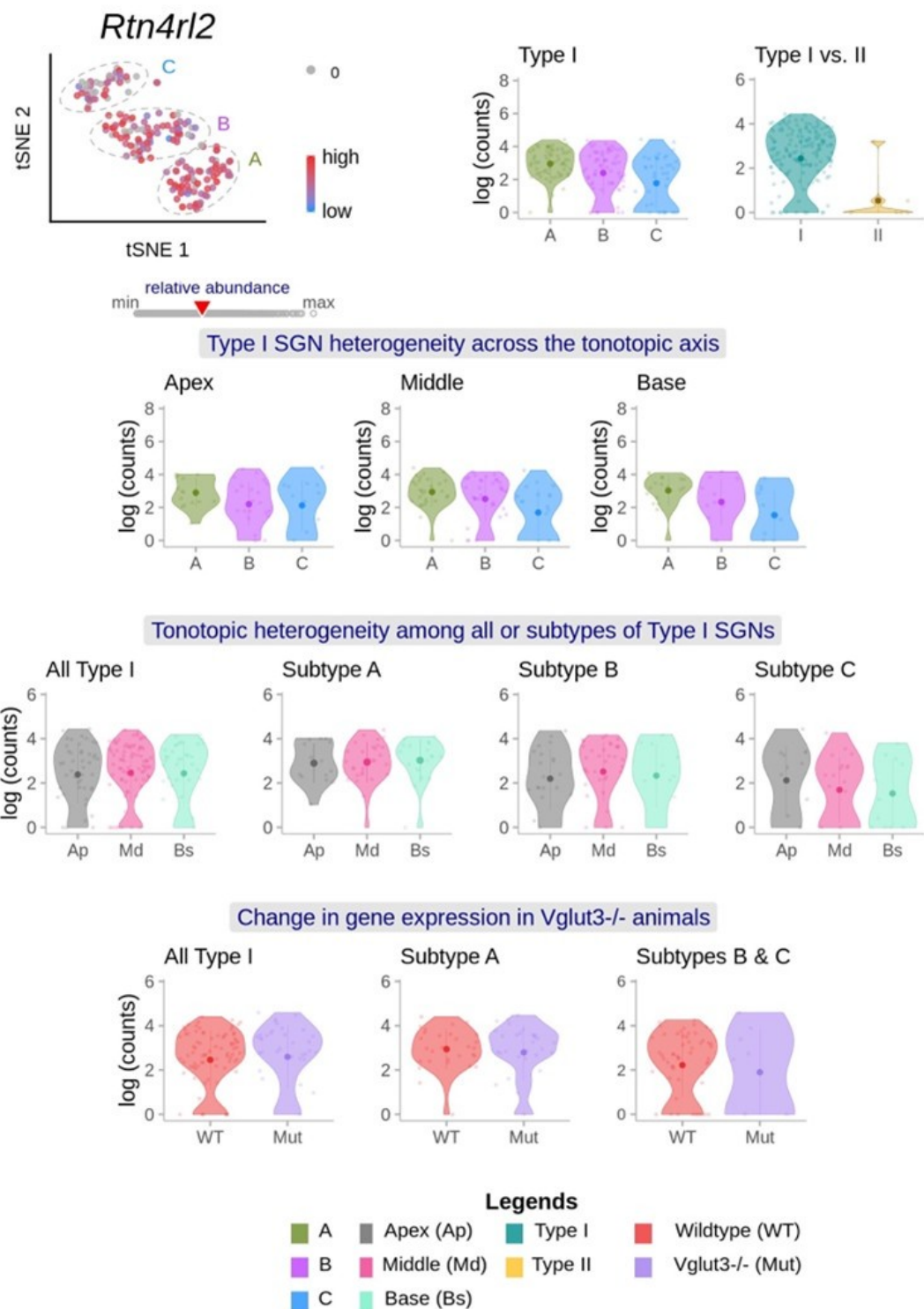
(2) It would be important to improve the RNAscope data. Controls should be provided for Figure 1B to show that no signal is observed in hair cells from knockouts. The authors apparently already have the sections because they analyzed gene expression in SGNs of the knock-outs (Figure 1C).

In Figure 1C gene expression in SGNs was assessed at p40, while the expression in hair cells is provided for p1 animals. Unfortunately, we do not have KO controls for p1 animals. However, as indicated in our manuscript, previously published RNA expression datasets do find RTN4RL2 expression in hair cells. Therefore, we think it is unlikely that our results are unspecific.

(3) It is unclear from the immunolocalization data in Figure 1D if all type I SGNs express RTN4RL2. Quantification would be important to properly document the presence of RTN4RL2 in all or a subset of type I SGNs. If only a subset of SGNs express RTN4RL2, it could significantly affect the interpretation of the data. For example, SGNs selectively projecting to the pillar or modiolar side of hair cells could be affected. These synapses significantly differ in their properties.

According to already published single cell RNAseq dataset from Shrestha et al., 2018, RTN4RL2 expression does not seem to show a clear type I SGN subtype specificity (Author response image 1). In response to the reviewer's comment, we have further performed anti-Parvalbumin (PV) and anti-calretinin (CR) immunostainings in mid-modiolar cryosections of RTN4RL2^{+/+} and RTN4RL2^{-/-} cochleae. Parvalbumin was chosen to label all SGNs and CALB2 was chosen primarily as a type Ia SGN marker (Sun et al., 2018). We present the data from all analyzed samples below (figure 2 of this rebuttal letter). Cell segmentation masks of PV positive cells were obtained using Cellpose 2.0 and the average CR intensity was calculated in those masks. While the distributions of CR intensity and the ratio of CR and PV intensities are slightly shifted in RTN4RL2^{-/-} cochleae, we take the data to suggest that the composition of the spiral ganglion by molecular type I SGN subtypes is largely unchanged in RTN4RL2^{-/-} mice.

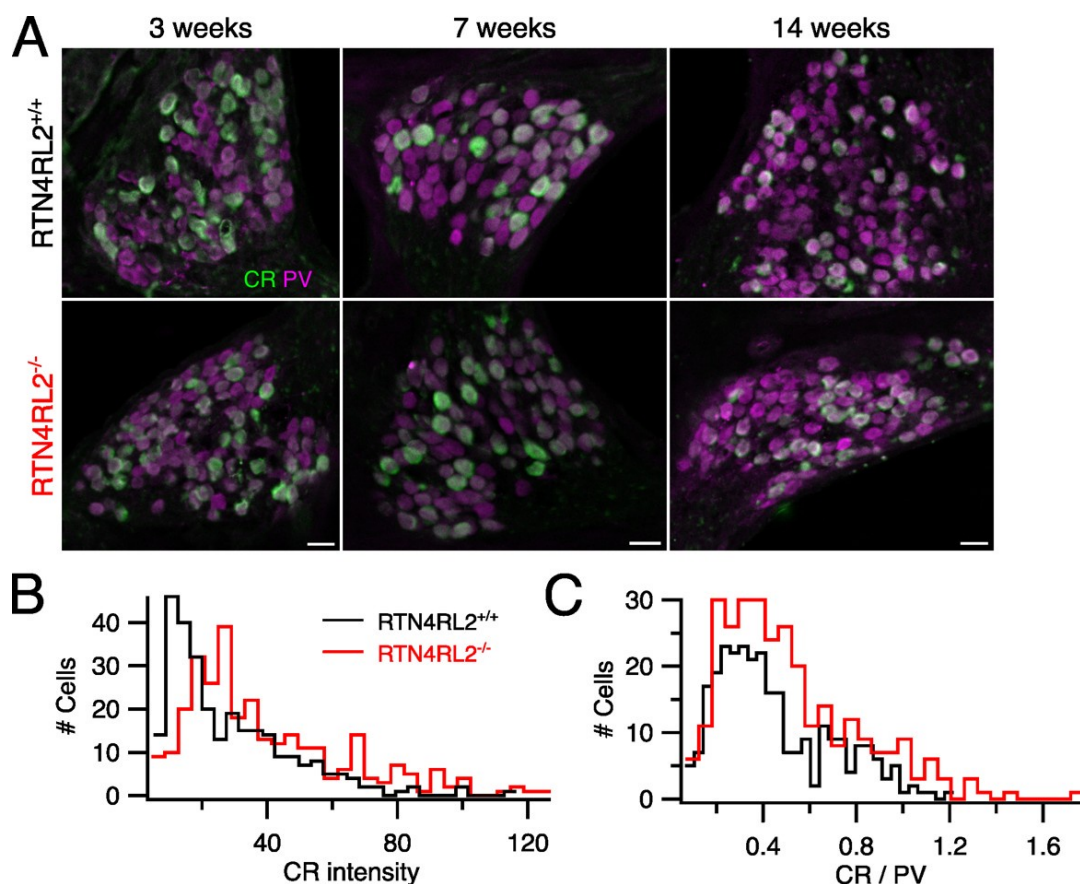
Author response image 1.



Author response image 1 cites single cell RNAseq data of Brikha R Shrestha, Chester Chia, Lorna Wu, Sharon G Kujawa, M Charles Liberman, Lisa V Goodrich. Sensory neuron diversity in the inner ear is shaped by activity. Cell. 2018 Aug 23; 174(5):1229-1246.e17. doi: 10.1016/j.cell.2018.07.007

Author response image 2.

Calretinin intensity distribution in spiral ganglion of RTN4RL2^{+/+} and RTN4RL2^{-/-} mice. (A) Mid-modiolar cochlear cryosections from RTN4RL2^{+/+} (top) and RTN4RL2^{-/-} (bottom) mice immunolabeled against Parvalbumin (PV) and Calretinin (CR). Scale bar = 20 μ m. (B) Distribution of CR intensity in PV positive cells (N = 3 for each genotype). (C) Distribution of the ratio of CR and PV intensities (N = 3 for each genotype).



(4) It is important to show proper controls for the RTN4RL2 immunolocalization data to show that no staining is observed in knockouts.

Unfortunately, our recent attempts to perform RTN4RL2 immunostainings on cryosections failed and therefore, we decided to remove the RTN4RL2 immunostainings from Figure 1. We have adjusted the results section accordingly.

(5) The authors state in the discussion that no staining for RTN4RL2 was observed at synaptic sites. This is surprising. Did the authors stain multiple ages? Was there perhaps transient expression during development? Or in axons indicative of a role in outgrowth, not synapse formation?

We thank the reviewer for the comment. We have now tried RTN4RL2 immunostainings on cryosections at several developmental stages, but unfortunately this time did not succeed to obtain reproducible and reliable results. Therefore, we decided to also remove the previous immunostainings from Figure 1. We have adjusted the results section as well as removed our statement of not detecting RTN4RL2 near the synaptic regions from the discussion.

(6) In Figure 2 it seems that images in mutants are brighter compared to wildtypes. Are exposure times equivalent? Is this a consistent result?

Yes, the samples were prepared in parallel, imaged and analyzed in the same manner.

No, we did not observe consistent differences in brightness and also did not find it in the exemplary images of figure 2.

(7) The number of synaptic ribbons for wildtype in Figure 2 is at 10/IHCs, and in Figure 2 Supplementary Figure 2 at 20/IHCs (20 is more like what is normally reported in the literature). The value for mutant similarly drastically varies between the two figures. This is a significant concern, especially because most differences that are reported in synaptic parameters between wild-type and mutants are far below a 2-fold difference.

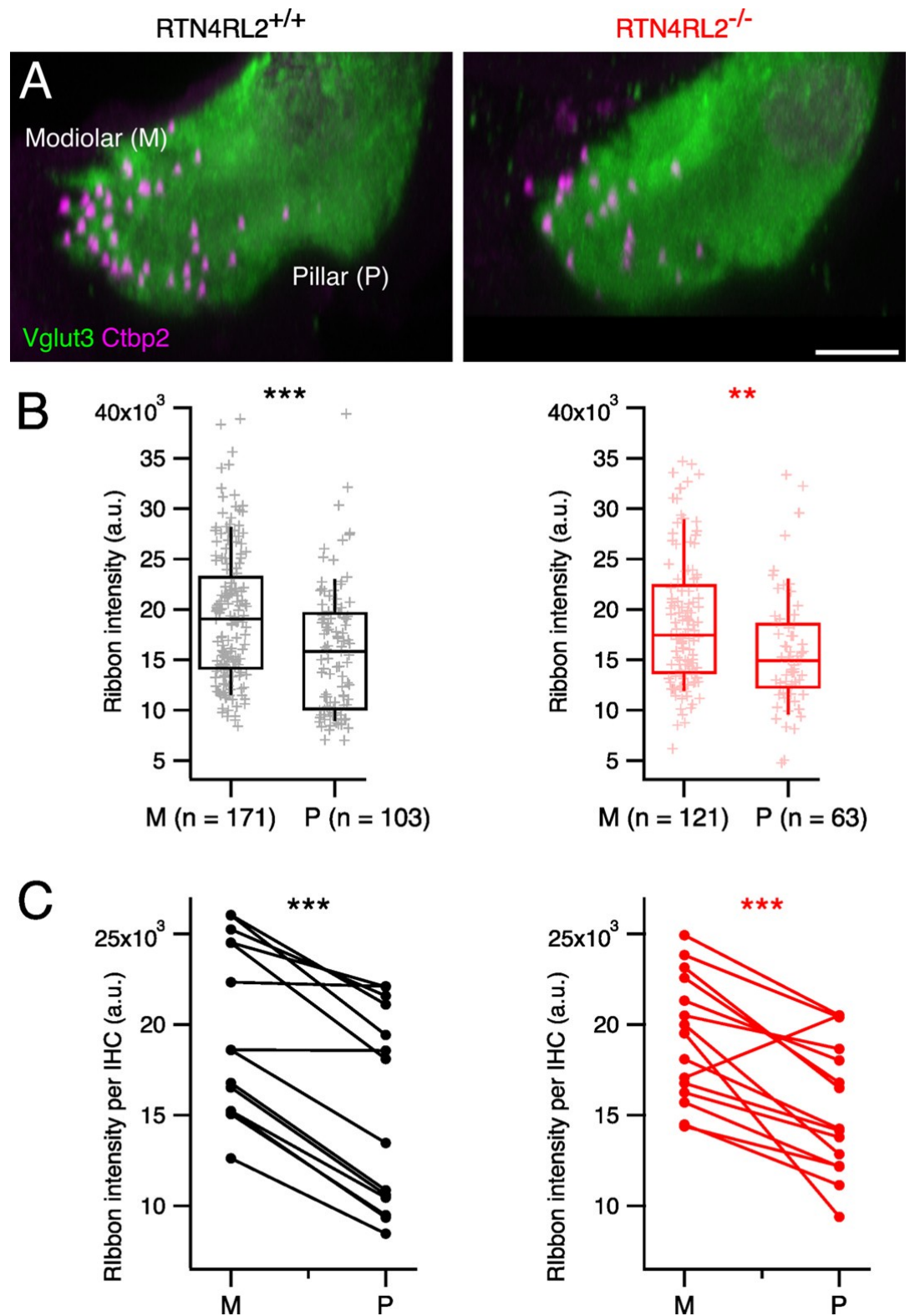
The key message is that there is no difference in the numbers of ribbons and synapses between the genotypes for the cochlear apex (~10 ribbons/IHCs, Figure 2 and Figure 2-figure supplement 2) and the mid- and base of the cochlea (more ribbons/IHCs, Figure 2-figure supplement 2). Figure 2-figure supplement 3 (now Figure 3) shows that there is a massive reduction of postsynaptic GluA2, while both Figure 2 and Figure 2-figure supplement 2 indicate that the number synapses is normal. These are two different data sets and while we closely collaborated and also shared the Moser lab protocols and analysis routines, we agree that there is a difference in the absolute synapse count, which most likely was an observer difference and different choice of tonotopic positions of analysis. In Figure 2 only the apical hair cells have been analyzed. The Moser lab, since establishing the immunofluorescence-based quantification of synapse number (Khimich et al., 2005) reported tonotopic differences in synapse counts (focus of Meyer et al., 2009 and reported by others: e.g. Kujawa and Liberman, 2009): apical and basal IHCs lower synapse numbers than mid-cochlear IHCs.

(8) The authors report differences in ribbon volume between wild-type and mutant. Was there a difference between the modiolar/pillar region of hair cells? It is known that synaptic size varies across the modiolar-pillar axis. Maybe smaller synapses are preferentially lost?

We thank the reviewer for the comment. Unfortunately, our already acquired datasets from 3-week-old mice did not allow us to check whether the previously described modiolar-pillar gradient of the ribbon size was collapsed in RTN4RL2^{-/-} mice due to the not so well-preserved morphology of the inner hair cells in our preparations. However, since the number of the ribbons is not changed in the RTN4RL2 KO mice, we do not think that the increase in the ribbon size is due to the loss of small ribbons. In response to the reviewers comment we have analyzed the modiolar-pillar gradient of the ribbon size in IHCs of middle turn of the cochlea from a newly acquired dataset of 14-week-old mice. We took the fluorescence intensity of Ctb2 positive puncta as a proxy for the ribbon size. In these older mice we found a preserved modiolar-pillar gradient of the ribbon size (larger ribbons at the modiolar side). We summarized the results in the below Author response image 3.

Author response image 3.

The modiolar-pillar gradient of ribbon size is preserved in RTN4RL2^{-/-} IHCs. (A) Maximum intensity projections of approximately 2 IHCs stained against Vglut3 and Ctb2 from 14-week-old RTN4RL2^{+/+} (left) and RTN4RL2^{-/-} (right) mice. Scale bar = 5 mm. (B) Synaptic ribbons on the modiolar side show higher fluorescence intensity than the ones on the pillar side of mid-cochlear IHCs in both RTN4RL2^{+/+} (left, N=2) RTN4RL2^{-/-} (right, N=2) mice. (C) Average fluorescence intensity of modiolar ribbons per IHC is higher than the average fluorescence intensity of pillar ribbons (paired t-test, p < 0.001).



(9) The authors show in Figure 2 - Supplement 3 that GluA2/3 staining is absent in the mutants. Are GluA4 receptors upregulated? Otherwise, synaptic transmission should be abolished, which would be a dramatic phenotype. Antibodies are available to analyze GluA4 expression, the experiment is thus feasible. Did the authors carry out recordings from SGNs?

In response to the reviewer's comment, we have performed GluA4 stainings in RTN4LR2^{-/-} mice and did not detect any GluA4 positive signal in the mutants (new Figure 3-figure supplement 1). Unfortunately, our animal breeding license was expired at the time we received the reviews and that is why our results are from 14-week-old animals. To verify that the absence of GluA4 signal is not due to potential PSD loss in 14-week-old RTN4LR2^{-/-}, we have additionally performed anti-Ctbp2, anti-Homer1 and anti-Vglut3 stainings in 14-week-old animals. Despite the reduced number, we still observed juxtaposing pre- and postsynaptic puncta. We assume that the reviewer asks for patch-clamp recordings from SGNs, which are, as we are confident the reviewer is aware of, technically very challenging and beyond the scope of the present study but an important objective for future studies. In response to the reviewers comment we have added a statement to the discussion pointing to these patch-clamp recordings from SGNs as important objective for future studies.

(10) The authors use SBEM to analyze SGN projections and synapses. The data suggest that a significant number of SGNs are not connected to IHCs. A reconstruction in Figure 3 shows hair cells and axons. It is not clear how the outline of hair cells was derived, but this should be indicated. Also, is this a defect in the formation of synapses and subsequent retraction of SGN projections? Or could RTN4LR2 mutants have a defect in axonal outgrowth and guidance that secondarily affects synapses? To address this question, it would be useful to sparsely label SGNs in mutants, for example with AAV vectors expression GFP, and to trace the axons during development. This would allow us to distinguish between models of RTN4LR2 function. As it stands, it is not clear that RTN4LR2 acts directly at synapses.

We agree with the reviewer on the value of a developmental study of afferent connectivity but consider this beyond the scope of the present study. In response to the reviewer's comment, we have replaced the IHC outlines with volume-reconstructed IHCs in Figure 3B (now Figure 4B). Moreover, as shown in Figure 3F (now Figure 4F), most if not all type-I SGNs (both with and without ribbon) were unbranched in the mutants just like in wildtype (also shown for a larger sample in Hua et al., 2021), arguing against morphological abnormality during development.

(11) The authors observe a tiny shift in the operation range of Ca²⁺ channels that has no effect on synaptic vesicle exocytosis. It seems very unlikely that this difference can explain the auditory phenotype of the mutant mice.

We assume that the statement refers to the normal exocytosis of mutant IHCs at the potential of maximal Ca²⁺ influx (Figure 3G and H, now Figure 4G and H). We would like to note that this experiment was performed to probe for a deficit of synapse function beyond that of the Ca²⁺ channel activation, but did not address the impact of the altered voltage—dependence of Ca²⁺ channel activation. In response to the reviewer's comment, we have now added further discussion to more clearly communicate that for the range of receptor potentials achieved near sound threshold we expect impaired IHC exocytosis as the Ca²⁺ channels require slightly more depolarization for activation in the mutant IHCs.

(12) ABR recordings were conducted in whole-body knockouts. Effects on auditory thresholds could be a secondary consequence of perturbation along the auditory pathway. Conditional knockouts or precisely designed rescue experiments would go a long way to support the authors' hypothesis. I realize that this is a big ask and floxed mice might not be available to conduct the study.

Thanks for this helpful comment and, indeed, unfortunately, we do not have conditional KO mice at our disposal. We totally agree that this will be important also for clarifying the role of

IHC vs. SGN expression of RTN4RL2. In response to the reviewer's comment, we now discussed the shortcoming of using constitutive RTN4RL2^{-/-} mice and added this important experiment on IHC and SGN specific deletion of RTN4RL2 as an objective of future studies.

Reviewer #3 (Public review):

In this study, the authors used RNAscope and immunostaining to confirm the expression of RTN4RL2 RNA and protein in hair cells and spiral ganglia. Through RTN4RL2 gene knockout mice, they demonstrated that the absence of RTN4RL2 leads to an increase in the size of presynaptic ribbons and a depolarized shift in the activation of calcium channels in inner hair cells. Additionally, they observed a reduction in GluA2/3 AMPA receptors in postsynaptic neurons and identified additional "orphan PSDs" not paired with presynaptic ribbons. These synaptic alterations ultimately resulted in an increased hearing threshold in mice, confirming that the RTN4RL2 gene is essential for normal hearing. These data are intriguing as they suggest that RTN4RL2 contributes to the proper formation and function of auditory afferent synapses and is critical for normal hearing. However, a thorough understanding of the known or postulated roles of RTN4RL2 is lacking.

We would like to thank the reviewer for appreciating the work and the advice that helped us to further improve the manuscript. We have carefully addressed all concerns, please see our point-per-point response below and the revised manuscript.

While the conclusions of this paper are generally well supported by the data, several aspects of the data analysis warrant further clarification and expansion.

(1) A quantitative assessment is necessary in Figure 1 when discussing RNA and protein expression. It would be beneficial to show that expression levels are quantitatively reduced in KO mice compared to wild-type mice. This suggestion also applies to Figure 2-supplement 3.D, which examines expression levels.

The processing of our control and KO samples for RNAscope was not strictly done in parallel and therefore we would like to refrain from quantitative comparison.

(2) In Figure 2, the authors present a morphological analysis of synapses and discuss the presence of "orphan PSDs." I agree that Homer1 not juxtaposed with Ctbp2 is increased in KO mice compared to the control group. However, in quantifying this, they opted to measure the number of Homer1 juxtaposed with Ctbp2 rather than directly quantifying the number of Homer1 not juxtaposed with Ctbp2. Quantifying the number of Homer1 not juxtaposed with Ctbp2 would more clearly represent "orphan PSDs" and provide stronger support for the discussion surrounding their presence.

We appreciate the reviewer's comment. We did not perform this analysis primarily because "orphan" Homer1 puncta, as seen in our immunostainings, are distributed away from hair cells in diverse morphologies and sizes. This makes distinguishing them from unspecific immunofluorescent spots—also present in wild-type samples—challenging. In response to the reviewer's request, we analyzed the number of "orphan" Homer1 puncta in our previously acquired RTN4RL2^{+/+} and RTN4RL2^{-/-} samples. Using the surface algorithm in Imaris software, we applied identical parameters across all samples to create surfaces for Homer1-positive puncta (total Homer1 puncta). We quantified "orphan" Homer1 puncta as the difference between total and ribbon-juxtaposing Homer1 puncta and normalized this number to the IHC count. Our results showed 4.3 vs. 26.8 "orphan" Homer1 puncta per IHC in RTN4RL2^{+/+} and RTN4RL2^{-/-} samples, respectively. We note that variations in acquired volumes between samples may introduce confounding effects.

(3) In Figure 2, Supplementary 3, the authors discuss GluA2/3 puncta reduction and note that Gria2 RNA expression remains unchanged. However, there is an issue with the lack of quantification for Gria2 RNA expression. Additionally, it is noted that RNA expression was measured at P4. While the timing for GluA2/3 puncta assessment is not specified, if it was assessed at 3 weeks old as in Figure 2's synaptic puncta analysis, it would be inappropriate to link Gria2 RNA expression with GluA2/3 protein expression at P4. If RNA and protein expression were assessed at P4, please indicate this timing for clarity.

GluA2/3 immunostainings were performed in 1 to 1.5-month-old animals. We apologize for not indicating this before and have now included it in Figure 3 legend. The processing of our control and KO samples for RNAscope was not strictly done in parallel and therefore we would like to refrain from quantitative comparison.

(4) In Figure 3, the authors indicate that RTN4RL2 deficiency reduces the number of type 1 SGNs connected to ribbons. Given that the number of ribbons remains unchanged (Figure 2), it is important to clearly explain the implications of this finding. It is already known that each type I SGN forms a single synaptic contact with a single IHC. The fact that the number of ribbons remains constant while additional "orphan PSDs" are present suggests that the overall number of SGNs might need to increase to account for these findings. An explanation addressing this would be helpful.

In Figure 3 (now Figure 4), we found additional type-1 SGNs that are unconnected to IHC, in good agreement with "orphan PSDs" observed under the light microscope. Indeed, we also confirmed monosynaptic, unbranched fiber morphology (Figure 3F, now Figure 4F). Together, these results imply about a 20% increase in the overall number of SGNs, which however we did not observe in SGN soma counting.

(5) In Figure 4F and 5Cii, could you clarify how voltage sensitivity (k) was calculated? Additionally, please provide an explanation for the values presented in millivolts (mV).

Voltage sensitivity (k) was calculated as the slope of the Boltzmann fit to the fractional

activation curves: $G/G_{max} = \frac{1}{1 + \exp\left[\frac{V_m - V_{half}}{k}\right]}$, Where G is conductance, G_{max} is the maximum

conductance, V_m is the membrane potential, V_{half} is the voltage corresponding to the half maximal activation of Ca^{2+} channels and k (slope of the curve) is the voltage sensitivity of Ca^{2+} channel activation. We have now added this to our Materials and Methods section.

(6) In Figure 6, the author measured the threshold of ABR at 2-4 months old. Since previous figures confirming synaptic morphology and function were all conducted on 3-week-old mice, it would be better to measure ABR at 3 weeks of age if possible.

ABR measurements for comparisons in a cohort of age-matched mice require fully developed individuals. 3 weeks is the minimum age that is regarded for a mature ear. However, variation in developmental differences among one litter is very frequent that affects normal hearing thresholds. From our own experience we do not regard the ear fully functional before 6 weeks of age. Then hearing thresholds are lowest indicating full functionality. Since the C57BL/6 background strain has a genetic defect in the Cadherin 23-coding gene (*Cdh23*) at the *ahl* locus of mouse chromosome 10 these mice exhibit early onset and progression of age-related hearing loss starting at 5–8 months (Hunter & Willott, 1987). Therefore, we chose a "safe" time window for stable and unaffected ABR recordings of 2-4 months to provide most representative data.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Please include information on the validation of all the antibodies used in this study, or reference the relevant work where the antibodies were previously validated.

In response to the reviewer's comment, we have now included a table listing all primary antibodies used in this study. Where possible, we provide references for knockout (KO) validation. Otherwise, we refer to the manufacturer's information, as provided in the respective datasheets.

(2) Figure 2 illustrates the pre- and postsynaptic changes observed in RTN4RL2 knockout (KO) mice. Please specify the age of the mice and the cochlear region depicted and analyzed in Figure 2.

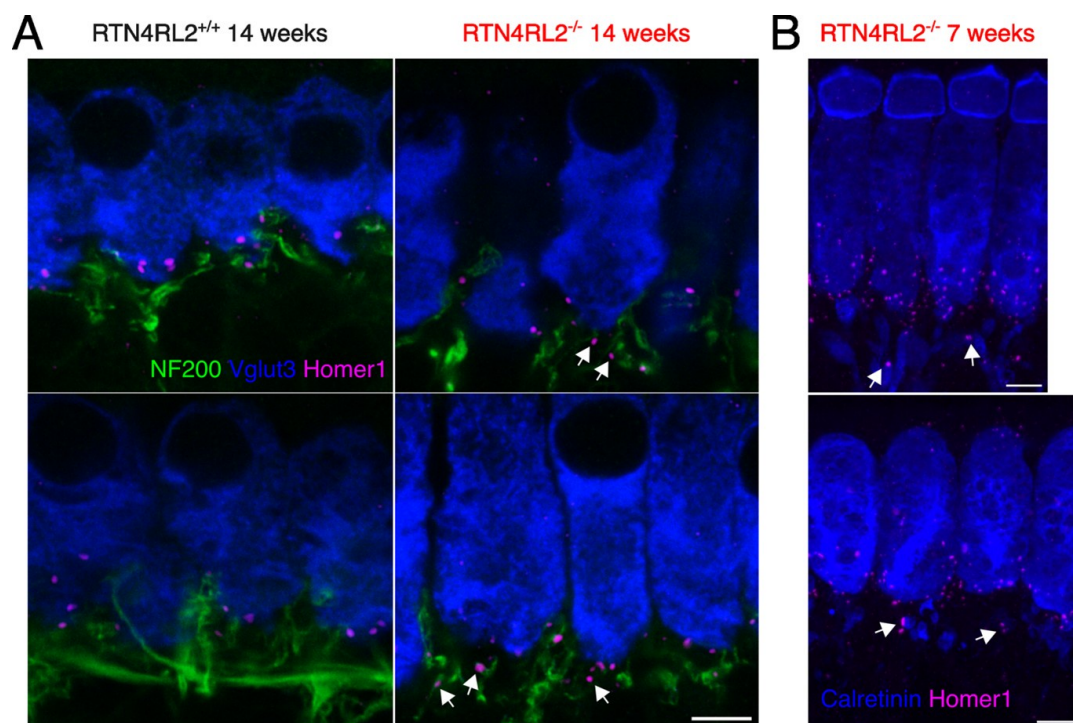
We thank the reviewer for the comment. The IHCs of apical cochlear region were analyzed in mice at 3 weeks of age. We have now added this to the figure legend.

(3) The discovery of orphan SGN neurites in RTN4RL2 KO mice is particularly intriguing. I wonder whether the additional Homer1-positive puncta illustrated in Figure 2 are present in these orphan SGN neurites, which would suggest that they may be functional. Conducting immunohistochemistry (IHC) labeling for type I SGN neurites using an anti-Tuj1 antibody, along with Homer1, would help localize the additional Homer1 puncta shown in Figure 2. Additionally, the "extra" Homer1 puncta appears less striking in the data presented in Figure 2-Supplement 2. Quantifying the number of Homer1 puncta in wild-type versus KO mice across different cochlear regions will help visualize the Figure 2-Supplement 2 data and relate the presence of extra neurites to the increased auditory brainstem response (ABR) thresholds observed at all frequencies.

We thank the reviewer for the comment and we agree that localizing orphan PSDs on the SGN neurites would be very useful. Unfortunately, the animal breeding license in the Göttingen lab had expired. At the time we received the reviews we only had access to 14-week-old animals and could not perform the stainings in animals which would have comparable age range to the rest of the study (3-4 weeks). The phenotype of extra Homer1 puncta was not as drastic in 14-week-old animals as it was in previously stained 3-week-old animals. Nevertheless, we still tried NF200, Homer1 and Vglut3 immunostainings in 14-week-old animals. We present representative single imaging planes of NF200, Homer1 and Vglut3 stainings in Author response image 4. Additionally, we provide exemplary images from 7-week-old RTN4RL2^{-/-}, where it looks like that the orphan Homer1 puncta are found on calretinin positive neurites.

Author response image 4.

Attempts to localize "orphan" Homer1 patches on type I SGN neurites. (A) Single exemplary imaging planes of apical IHC region from RTN4RL2^{+/+} (left) and RTN4RL2^{-/-} (right) mice immunolabeled against NF200, Vglut3 and Homer1. White arrows show putative "orphan" Homer1 puncta on NF200 positive neurites. Scale bar = 5 mm. (B) Maximum intensity projections of representative confocal stacks of IHCs from RTN4RL2^{-/-} mice immunolabeled against Calretinin and Homer1. Scale bars = 5 mm. White arrows show possible "orphan" Homer1 puncta on Calretinin positive boutons.



(4) The authors noted a reduction in the number of GluA2/3-positive puncta in RTN4RL2 KO mice, as shown in Figure 2-Supplement 3. However, in the Results section (page 5, line 124), it is unclear whether the authors refer to a reduction in fluorescence intensity or the number of puncta. Please clarify this.

We thank the reviewer for the comment. We refer to the number and have now added this to the manuscript.

(5) I find it particularly interesting that, despite the presence of smaller but synaptically engaged Homer1-positive SGN neurites, these appear to lack or present a reduction in the number of GluA2/3 puncta, and that GluA2/3 puncta are observed in non-ribbon juxtaposed neurites. Therefore, I suggest including GluA2/3 (Fig2 supplement 3) data in the main figure. It would be valuable to determine whether the orphan neurites express both Homer1 and GluA2/3, which could indicate that the defect is not solely due to reduced GluA2/3 expression at the formed synapses, but also to the presence of additional orphan synapses. I would also mention in the discussion how the phenotype of the RTN4RL2 KO compares to the GluA2/3 KO and if the lack of GluA2/3 at the AZ could explain the increase in ABR threshold. Quantification of GluA2/3 puncta at the apical, middle, and basal region would also help understand the auditory phenotype of the KO mice.

We have changed Figure2-figure supplement 3 to become a main figure (Figure 3) based on the recommendation of the reviewer. We agree, that it would be valuable to perform immunohistochemistry combining anti-GluA2/3 and anti-Homer1 and anti-Ctbp2 antibodies to see if the “orphan” Homer1 patches house GluA2/3 not juxtaposing synaptic ribbons. Unfortunately, as mentioned above, due to the expiration of our animal breeding and experimentation licenses we did not manage to do those experiments. We have however performed stainings with anti-GluA4 antibodies and could not detect GluA4 signal in RTN4RL2^{-/-} mice (Figure 3-figure supplement 1). This potentially could explain the more

drastic ABR threshold elevation in RTN4RL2^{-/-} mice compared to e.g. GluA3 KO mice. We have now made this clearer in our discussion.

(6) I suggest considering the use of color-blind friendly palettes for figures and graphs in this manuscript to enhance clarity and ensure that the findings are accessible to a wider audience and improve the overall effectiveness of the presentation. Please use color-blind-friendly schemes in Figure 1 and Figure 2 Supplement 3.

Done.

(7) Could you please explain what "XX {plus minus} Y, SD = W" means in the figure legends?

Mean ± SEM (standard error of the mean), SD (standard deviation) are indicated in the legends. In response to the reviewer comment we have now added an explanation in the Materials and Methods → Data analysis and statistics section.

(8) Please include information about the ear tested (left or right or both).

Both ears were tested. Since there was no significant difference between right and left ear we did not further consider this factor. We will add this fact more precisely in the Material and methods section.

Reviewer #3 (Recommendations for the authors):

(1) Line 90: Why not show this control, it is a nice control.

Unfortunately, our recent attempts to perform RTN4RL2 immunostaining on cryosections were unsuccessful. Therefore, we decided to remove RTN4RL2 immunostaining from Figure 1 and have adjusted the results section accordingly.

(2) Line 94: Please provide a reference for these interactions.

Done.

<https://doi.org/10.7554/eLife.103481.2.sa0>